SaaS universal wallet for coupons and rewards integrated with vendor nodes

CS 313 Section 1

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1. Introduction

Nowadays, businesses issue a wide variety of coupons, loyalty points, and rewards to attract and retain customers. However, this often results in a highly fragmented ecosystem where customers need to manage numerous physical cards, multiple mobile apps, or paper coupons. For customers, this leads to lost rewards and general inconvenience, while small businesses and franchise operations struggle to maintain engagement, track manual coupons accurately, or seamlessly handle cross-branch financial settlements. These inefficiencies directly impact customer retention and limit a business's ability to make data-driven decisions.

In this regard, a **Universal SaaS Digital Wallet and Reward Platform** has been proposed to address these challenges through a centralized, scalable digital ecosystem. The system is designed to bridge the gap between vendors, franchise branches, and consumers by offering a single universal wallet. It enables multi-branch businesses and independent vendors to deploy customizable loyalty campaigns, while automating the tracking, distribution, and cross-branch redemption of rewards without financial disputes or fraud risks.

The platform includes a structured interface that organizes the workflow into distinct, user-specific portals and key operational modules. Some of these essential modules include: the Customer Digital Wallet, Vendor Campaign Management, a Cross-Branch Reward Settlement System, and a secure QR Code Identification System. Through these modules, users can seamlessly collect points, vendors can deploy target marketing, and franchise networks can securely monitor multi-branch operations.

Through this system, it is possible to eliminate fragmented reward systems, protect businesses against duplicate coupon redemptions, and foster stronger, data-backed customer loyalty.

2. Design Goals

The design of the **Universal SaaS Digital Wallet and Reward Platform** focuses on simplifying loyalty management for both businesses and consumers while ensuring high scalability for franchise operations. The main design goals of the system include the following:

- **Unified Customer Wallet Experience**

The system is designed to allow customers to easily manage their personal rewards through a secure interface. It provides real-time access to available coupons, point balances, and

interactive digital stamp cards. Customers can quickly browse participating vendors, view their redemption history, and receive notifications regarding upcoming rewards and promotions.

- **Flexible Vendor Campaign Management**

The Vendor Management module enables business owners to register accounts, manage company profiles, and establish distinct marketing initiatives. It provides comprehensive functions for configuring specific reward conditions, allowing vendors to choose between **Point-based systems or Digital Stamp campaigns** to best fit their business model.

- **Fraud-Resistant QR Code Verification**

The QR Code module automates the check-out workflow by serving as a secure tool for customer identification. By allowing vendor staff to scan customer-presented QR codes, the system ensures that awarding points/stamps and verifying coupon redemptions happens instantly, eliminating manual entry errors and eliminating duplicate redemption fraud.

- **Seamless Multi-Branch and Cross-Branch Synchronization**

The platform is engineered to effortlessly scale across multi-branch and complex franchise operations. It enables vendors to manage multiple branches under a single corporate profile, tracking rewards earned at one location and redeemed at another to maintain accounting clarity and prevent operational disputes.

- **Comprehensive Administrative Governance**

The system features a dedicated Administrator portal designed to maintain the integrity and security of the entire SaaS platform. It equips platform admins with the tools necessary to **verify vendor registrations, approve or reject business accounts, manage all system users, and suspend fraudulent accounts** whenever suspicious behavior is detected.

- **Data-Driven Business Insights and System Analytics**

The platform provides robust reporting structures tailored to different user roles. Vendors gain access to customer engagement reports and redemption histories to optimize their campaigns, while Platform Administrators are equipped with global system activity monitoring and high-level analytics to maintain overall platform health and configuration.

- **Role-based Access control**

The platform provides role-based access for Customers, Vendors, and Administrators, ensuring that each user can only access features and data relevant to their responsibilities. This improves security, simplifies navigation, and protects sensitive information.

3. System Behaviors

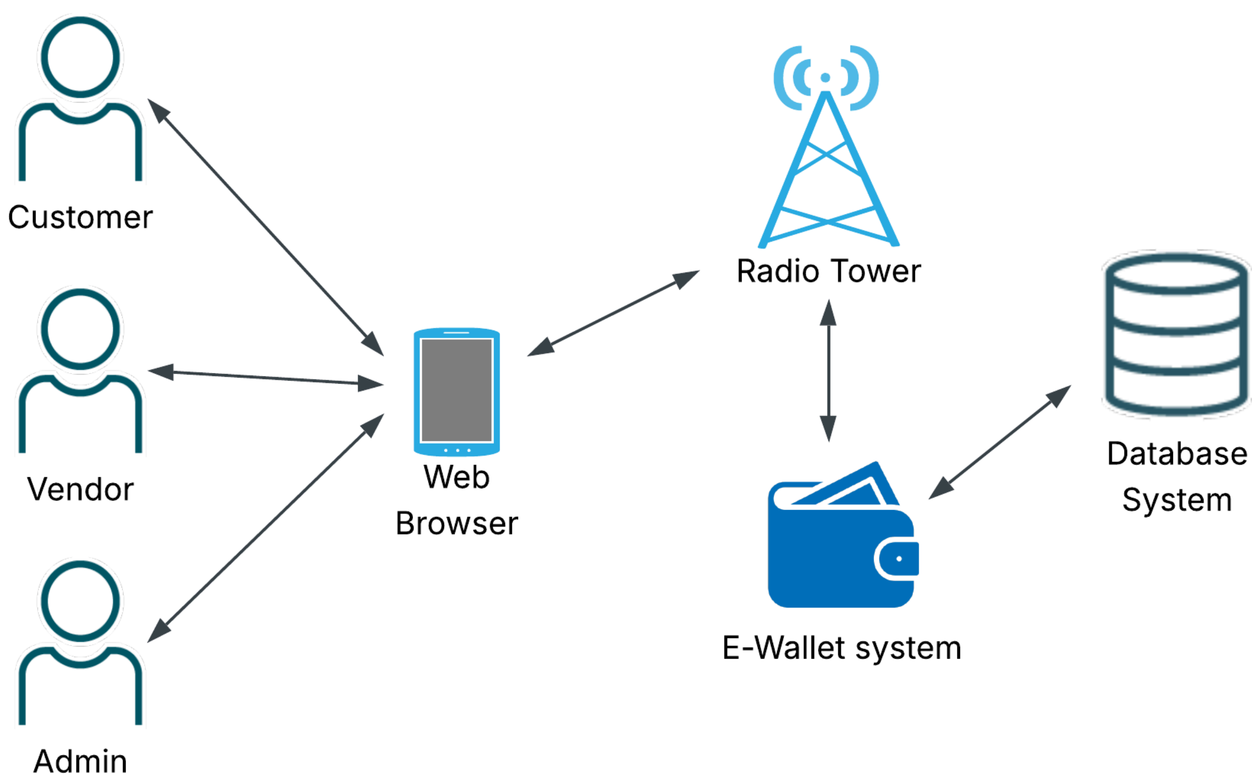
The **Universal SaaS Wallet and Rewards Platfor**m supports interactions between Customers, Vendors, and Administrators through a set of coordinated business processes. Users authenticate securely using their Google accounts before accessing the platform. After authentication, customers can manage digital wallets, earn and redeem rewards, and use QR codes for transactions. Vendors manage campaigns, branches, and customer rewards, while administrators oversee vendor verification, user management, and platform monitoring. These behaviors ensure secure, efficient, and scalable platform operations.

3.1 Overview

The system follows a role-based workflow in which each user accesses features according to their assigned role after signing in with a Google account. Customers interact with digital wallets, rewards, and QR codes, Vendors manage campaigns and branches, and Administrators oversee platform governance and monitoring. The following architectural views illustrate how these system behaviors are organized and implemented.

4. Logical View

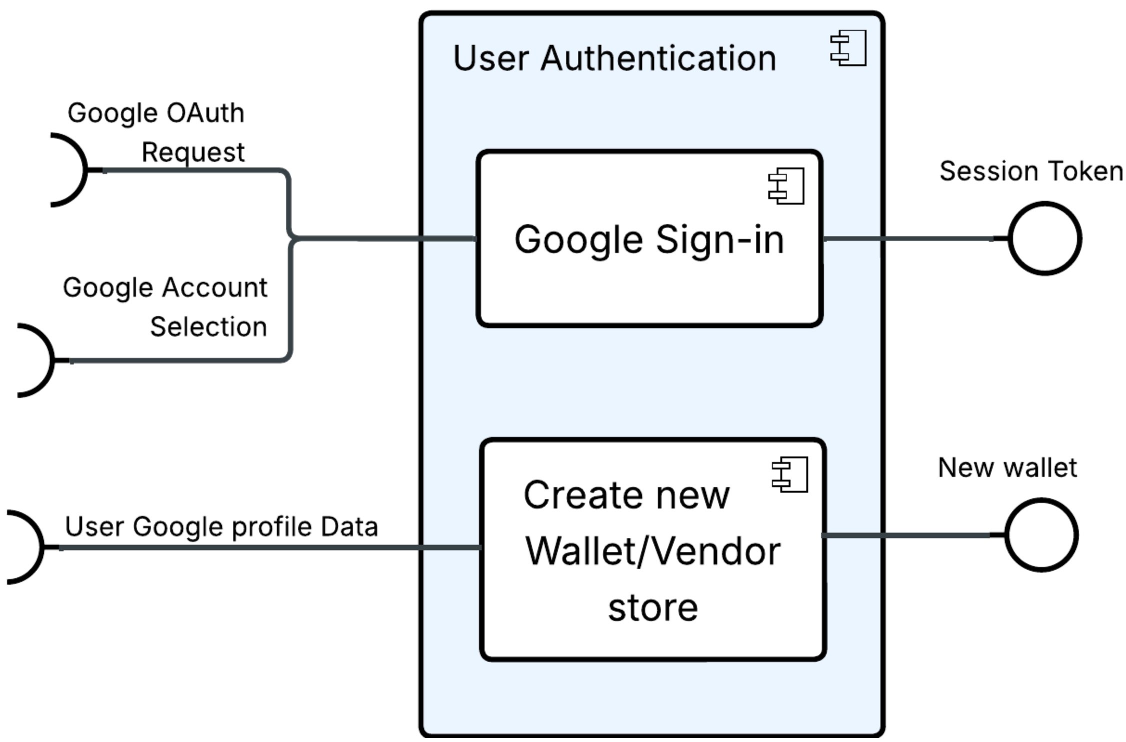
Figure 4.1 High-Level Design

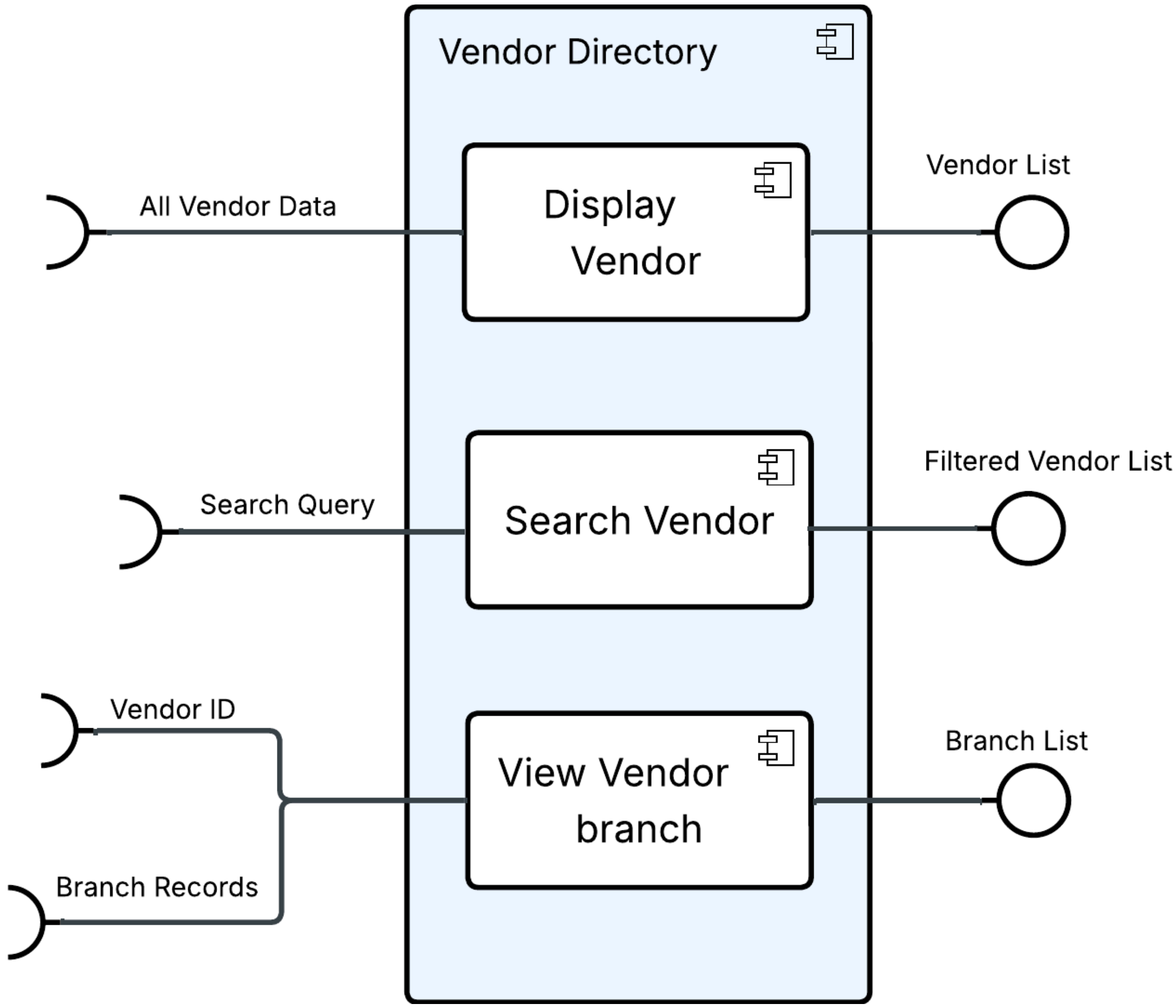


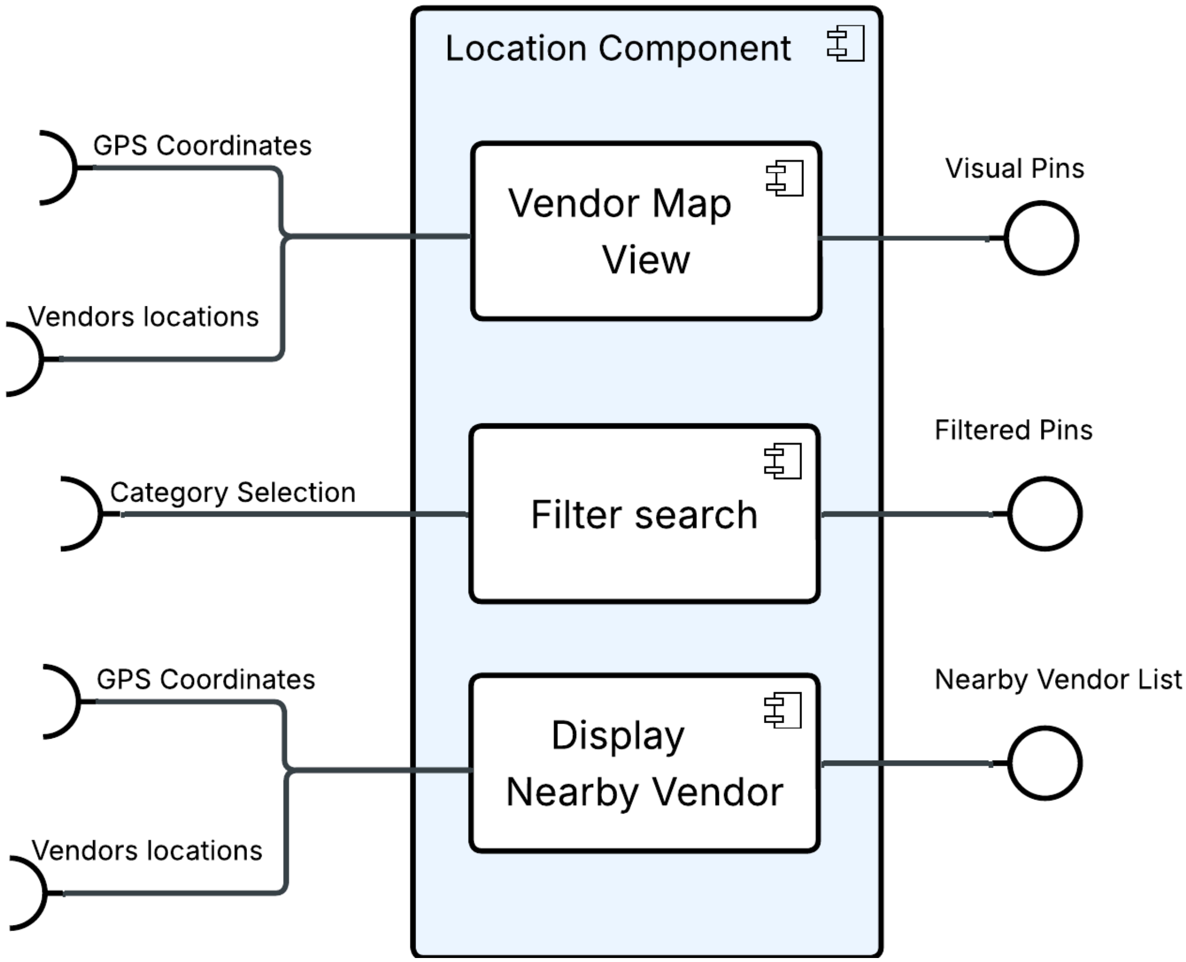
The High-Level Design presents the overall architecture of the Universal SaaS Wallet & Rewards Platform, showing the major system components and how they interact to support platform functionality.

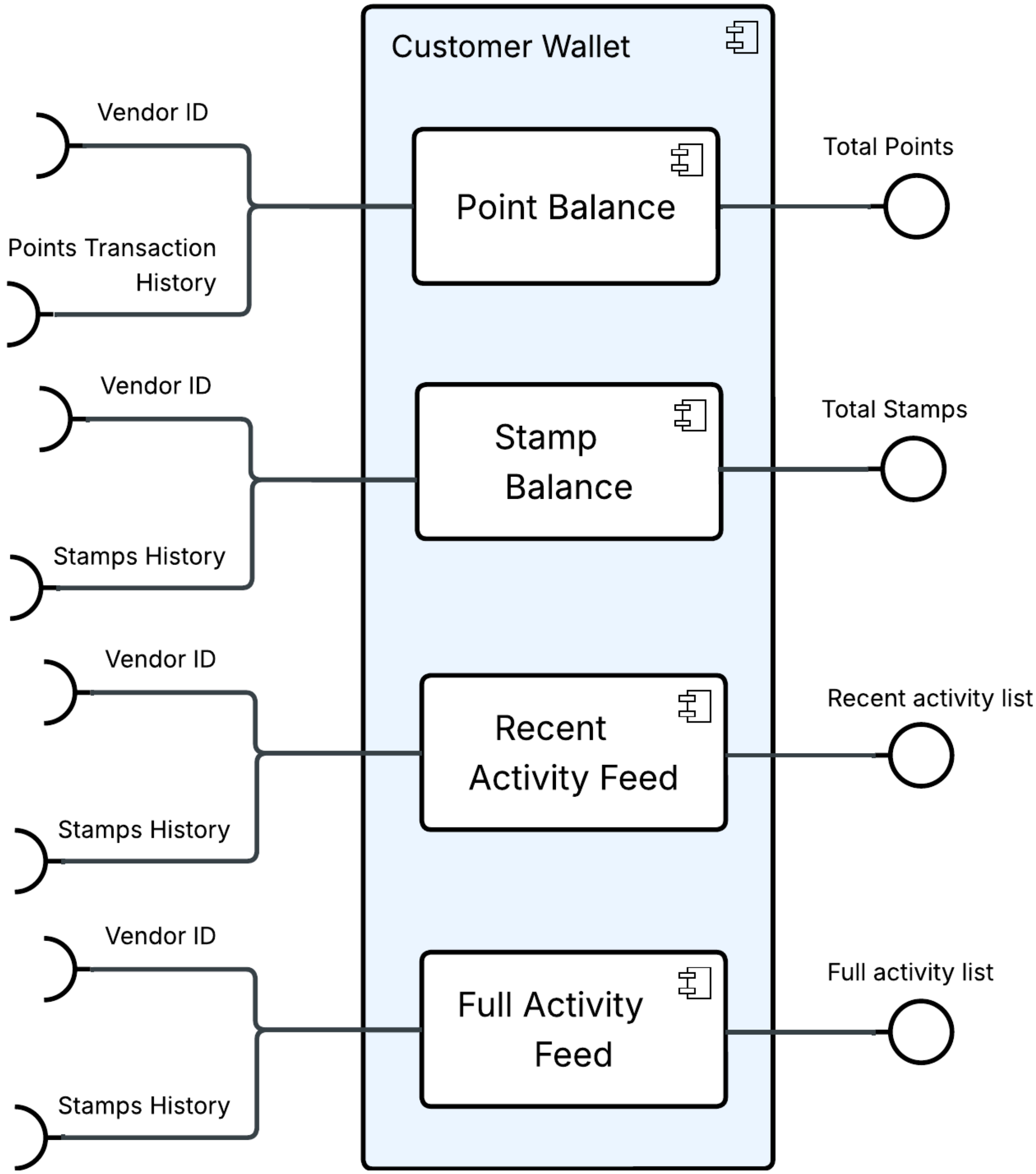
Figure 4.2 Mid-Level Design

- Customer

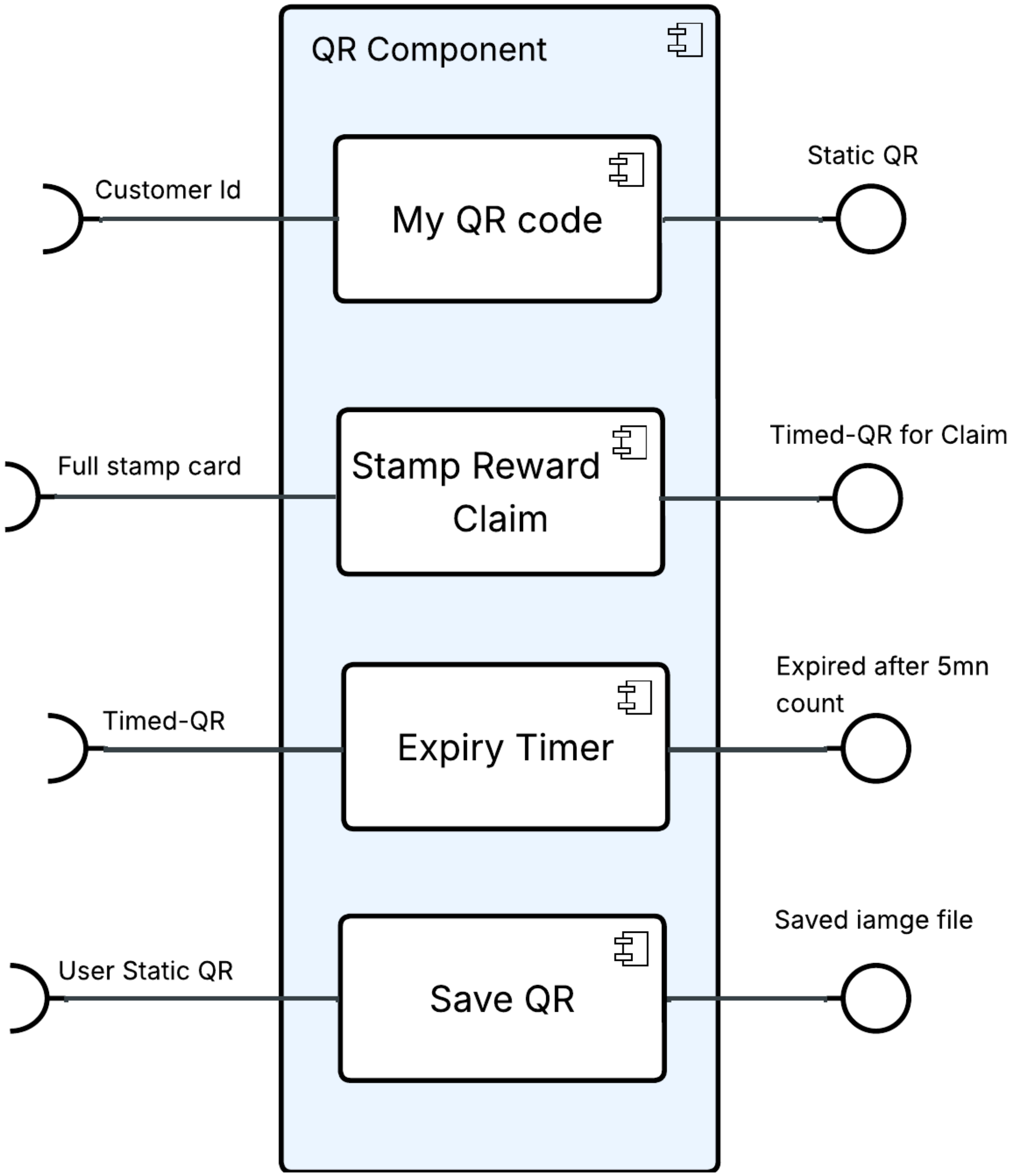


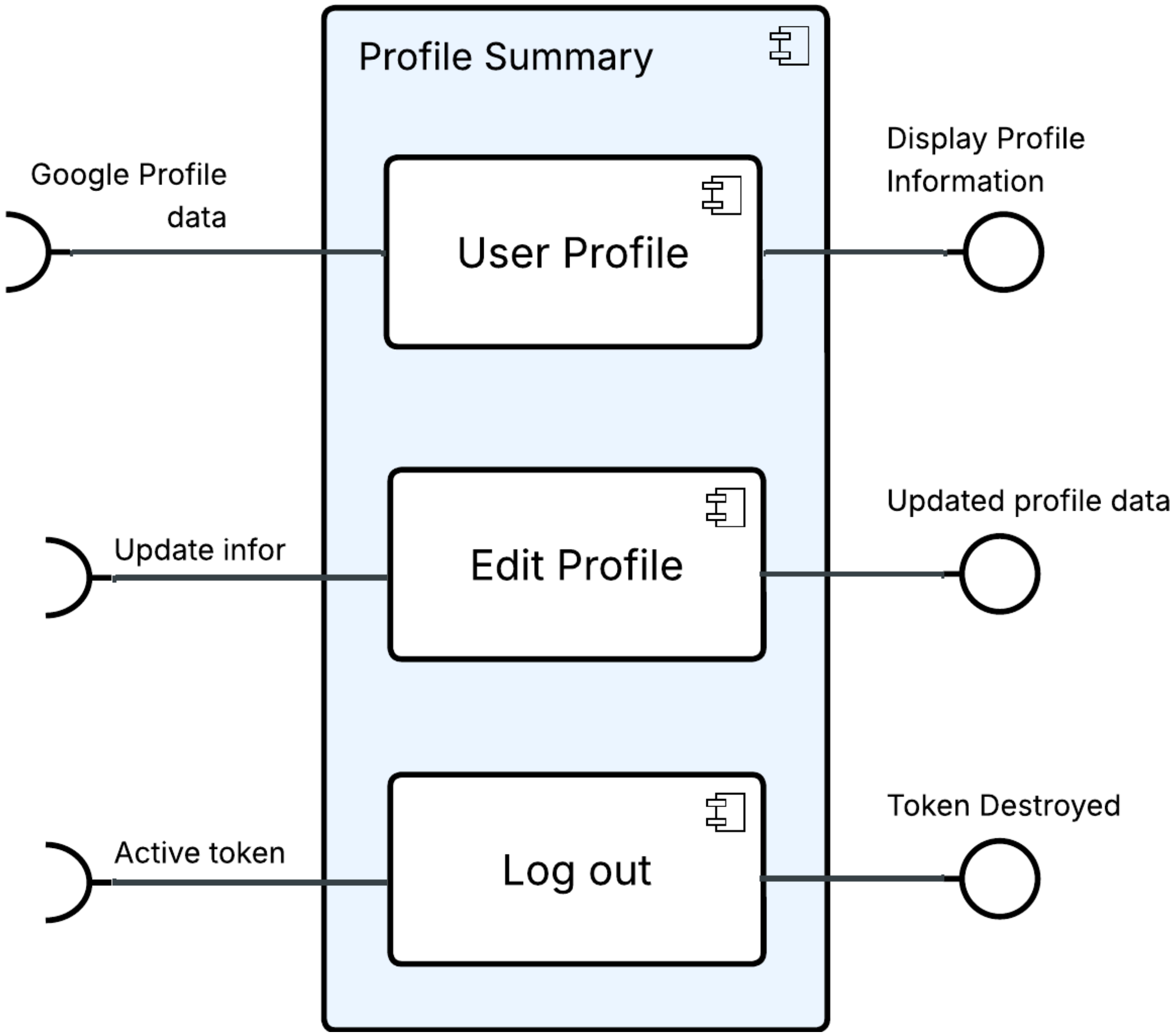

Vendor Directory
All Vendor Data
Display Vendor
Vendor List
Search Query
Search Vendor
Filtered Vendor List
Vendor ID
View Vendor branch
Branch List
Branch Records


Location Component
GPS Coordinates
Vendors locations
Vendor Map View
Visual Pins
Category Selection
Filter search
Filtered Pins
GPS Coordinates
Vendors locations
Display Nearby Vendor
Nearby Vendor List

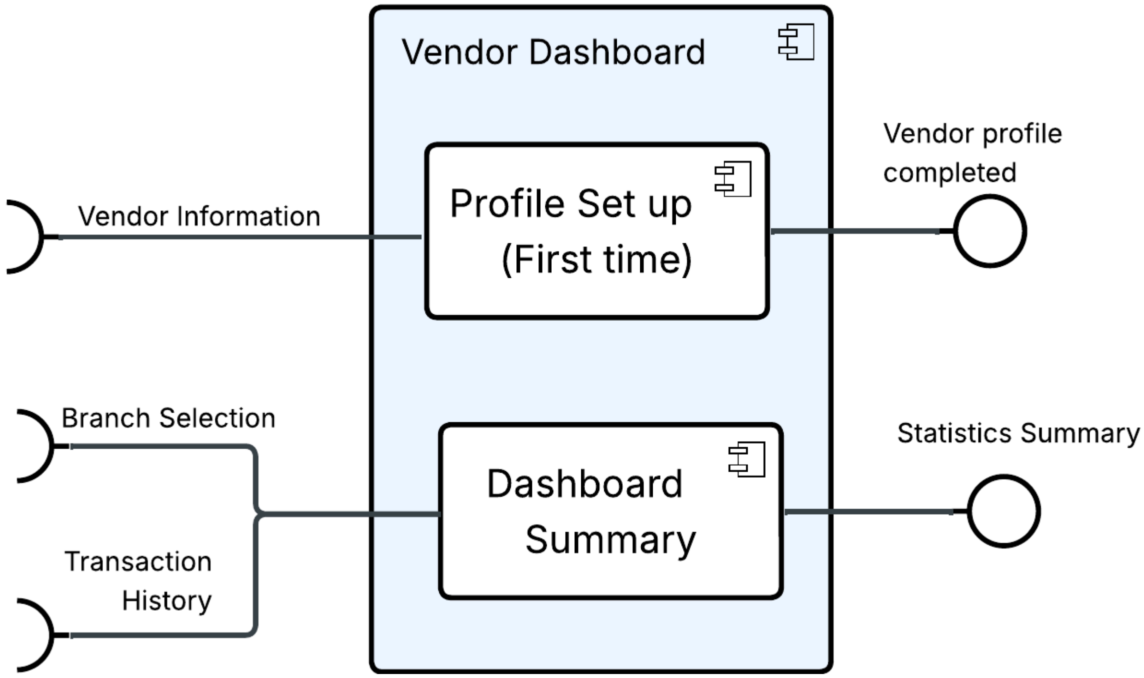


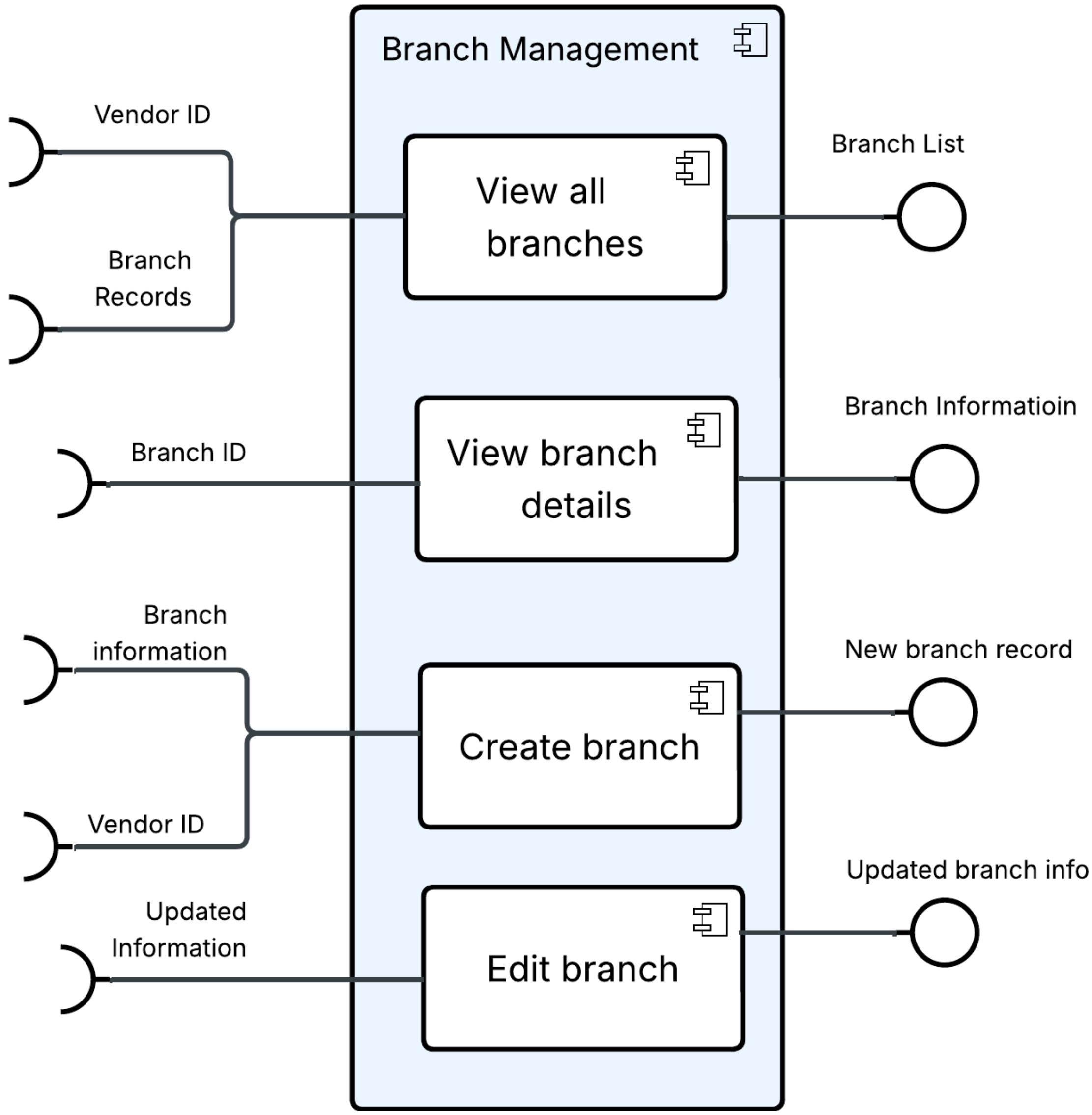
Customer Wallet
Vendor ID
Points Transaction History
Point Balance
Total Points
Vendor ID
Stamps History
Stamp Balance
Total Stamps
Vendor ID
Stamps History
Recent Activity Feed
Recent activity list
Vendor ID
Stamps History
Full Activity Feed
Full activity list

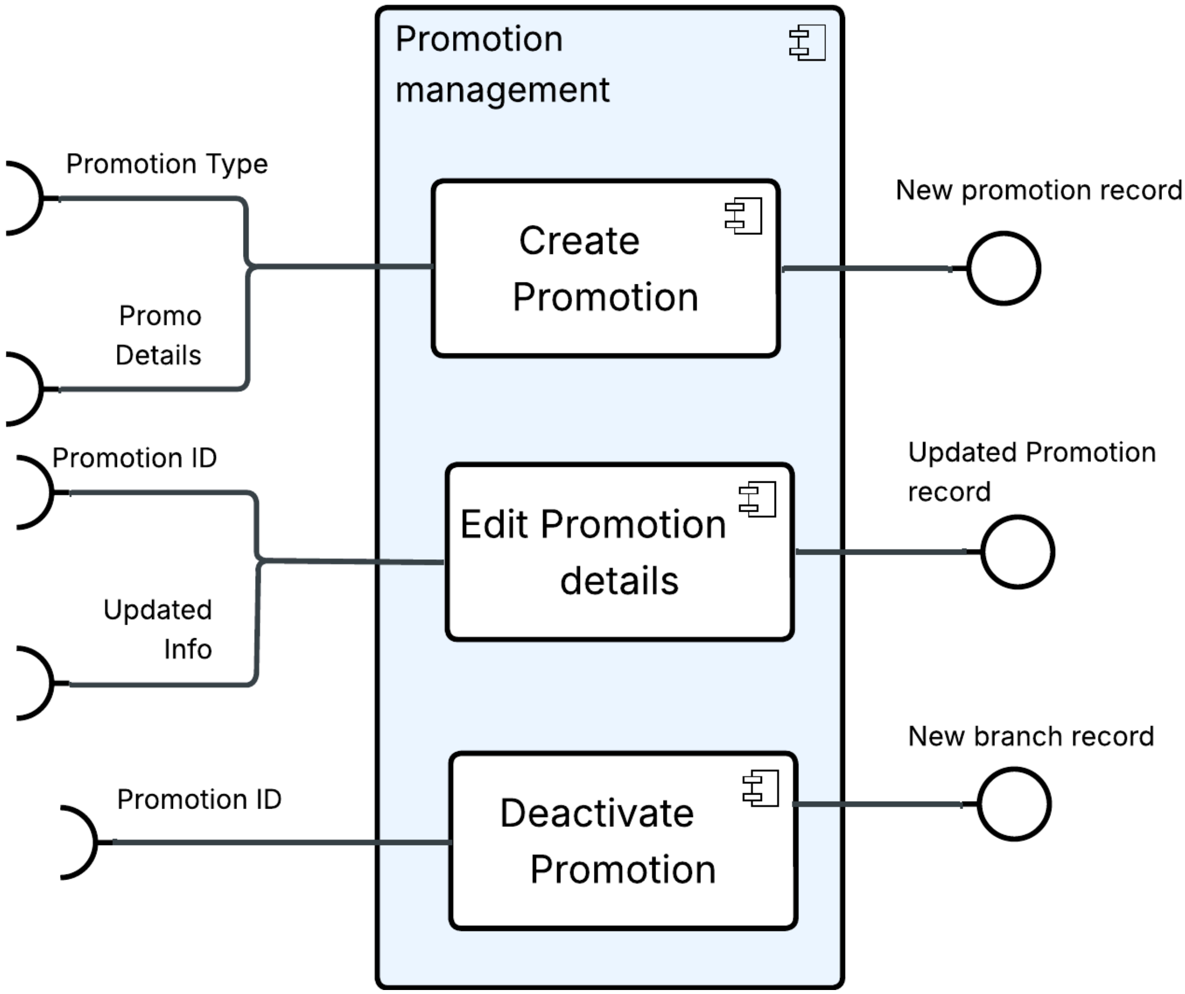

QR Component
Customer Id
My QR code
Static QR
Full stamp card
Stamp Reward Claim
Timed-QR for Claim
Timed-QR
Expiry Timer
Expired after 5mn count
User Static QR
Save QR
Saved iamge file

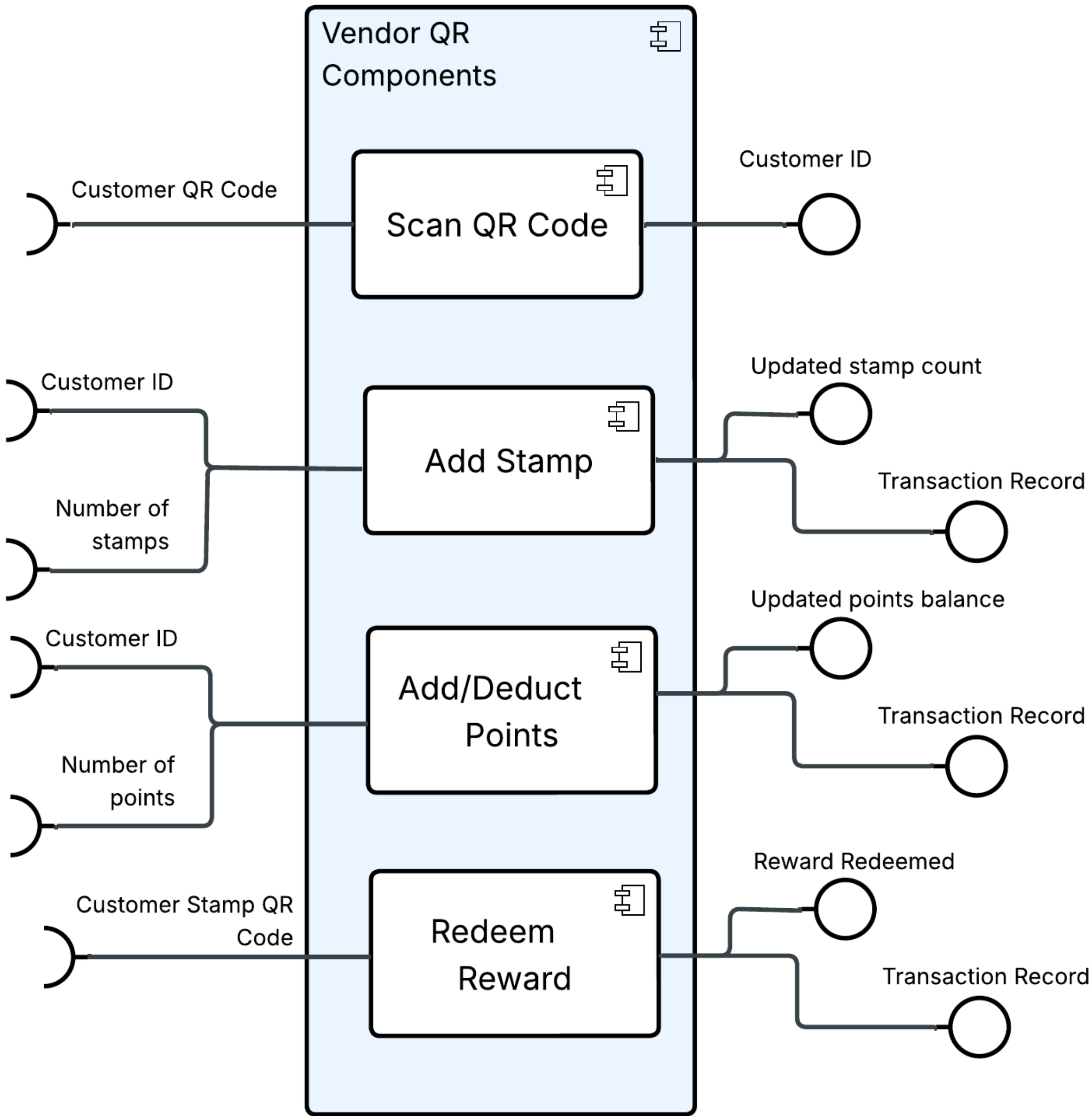

Profile Summary
Google Profile data
User Profile
Display Profile Information
Update infor
Edit Profile
Updated profile data
Active token
Log out
Token Destroyed

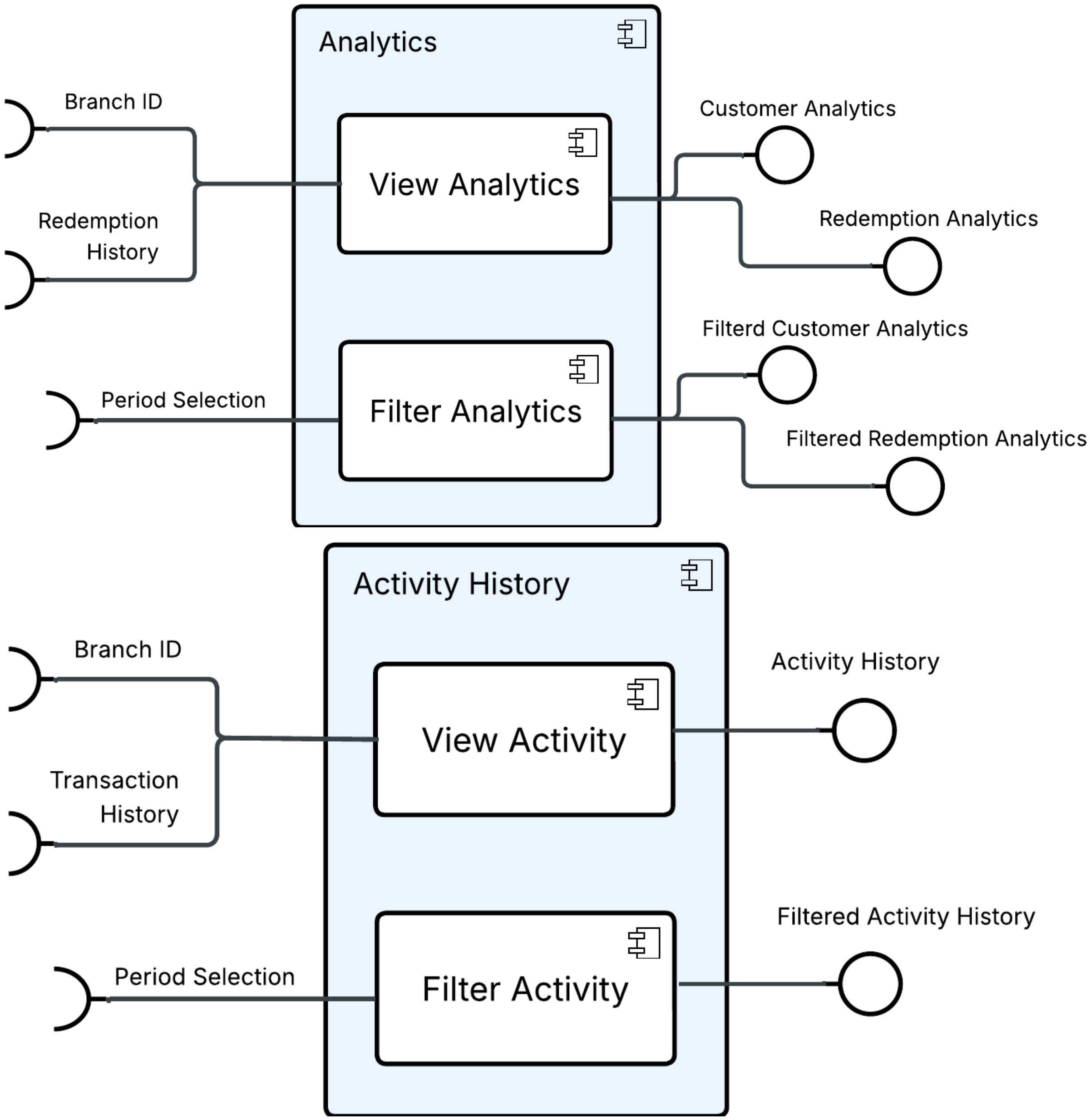
- Vendor


Vendor Dashboard
Profile Set up (First time)
Dashboard Summary
Vendor Information
Vendor profile completed
Branch Selection
Transaction History
Statistics Summary


Branch Management
Vendor ID
Branch
Records
View all
branches
Branch List
Branch ID
View branch
details
Branch Informatioin
Branch
information
Create branch
New branch record
Vendor ID
Updated
Information
Edit branch
Updated branch info


Promotion management
Promotion Type
Promo Details
Create Promotion
New promotion record
Promotion ID
Updated Info
Edit Promotion details
Updated Promotion record
Promotion ID
Deactivate Promotion
New branch record

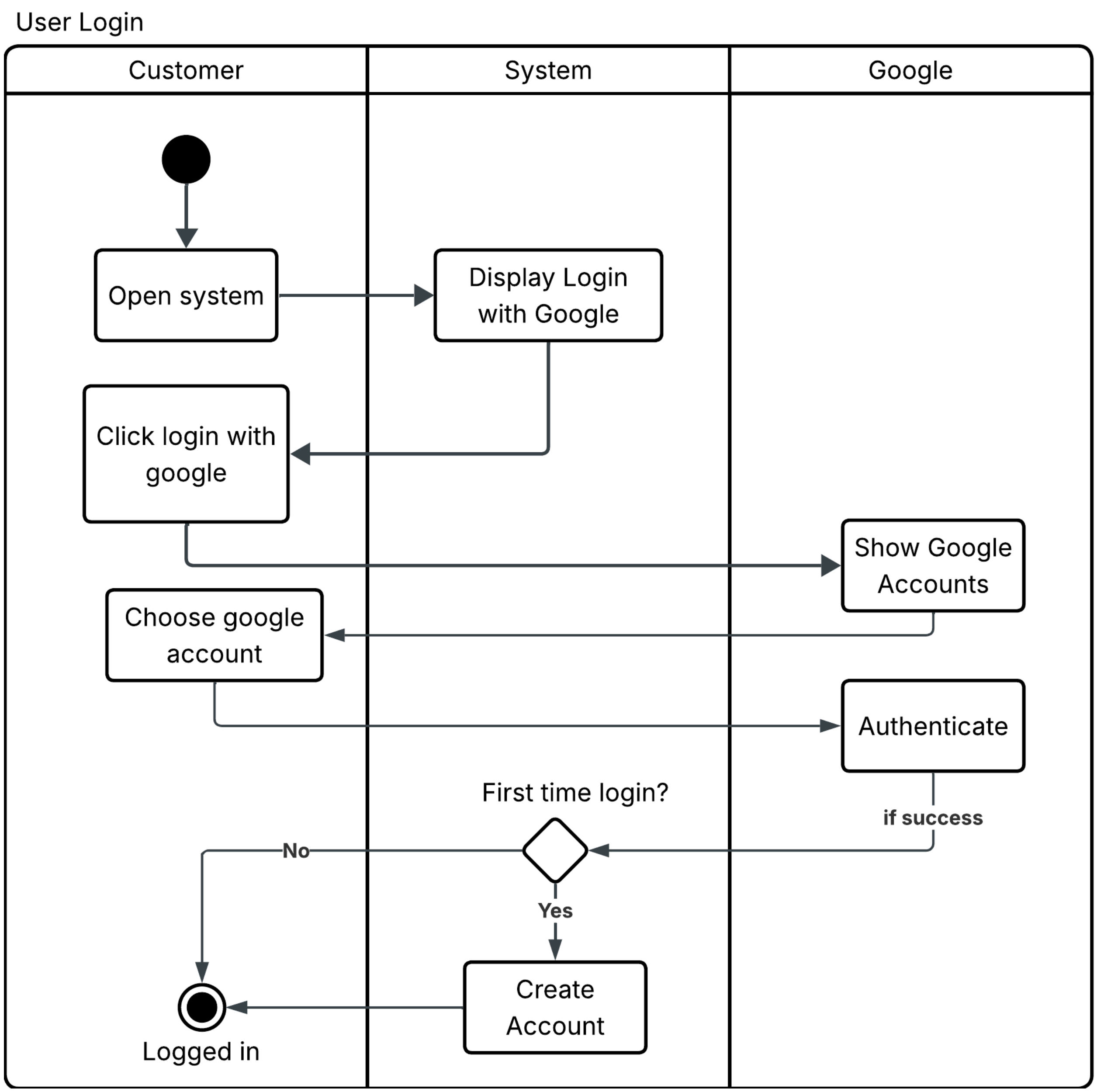

Vendor QR Components
Customer QR Code
Scan QR Code
Customer ID
Customer ID
Number of stamps
Add Stamp
Updated stamp count
Transaction Record
Customer ID
Number of points
Add/Deduct Points
Updated points balance
Transaction Record
Customer Stamp QR Code
Redeem Reward
Reward Redeemed
Transaction Record



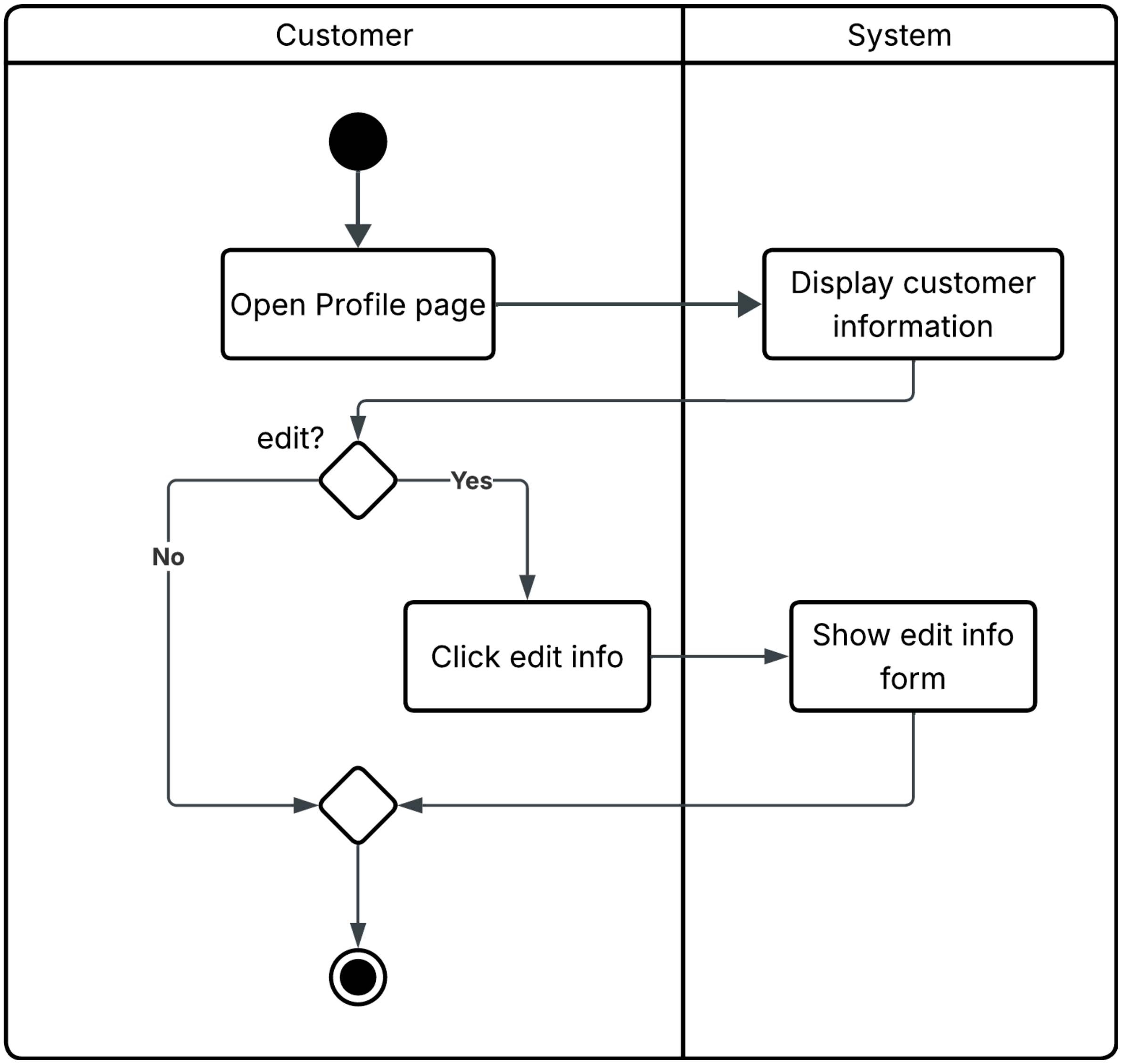
The Mid-Level Design provides a more detailed view of the platform by decomposing the major components into functional modules. Each module is responsible for a specific business capability while collaborating with other modules to support the overall system.

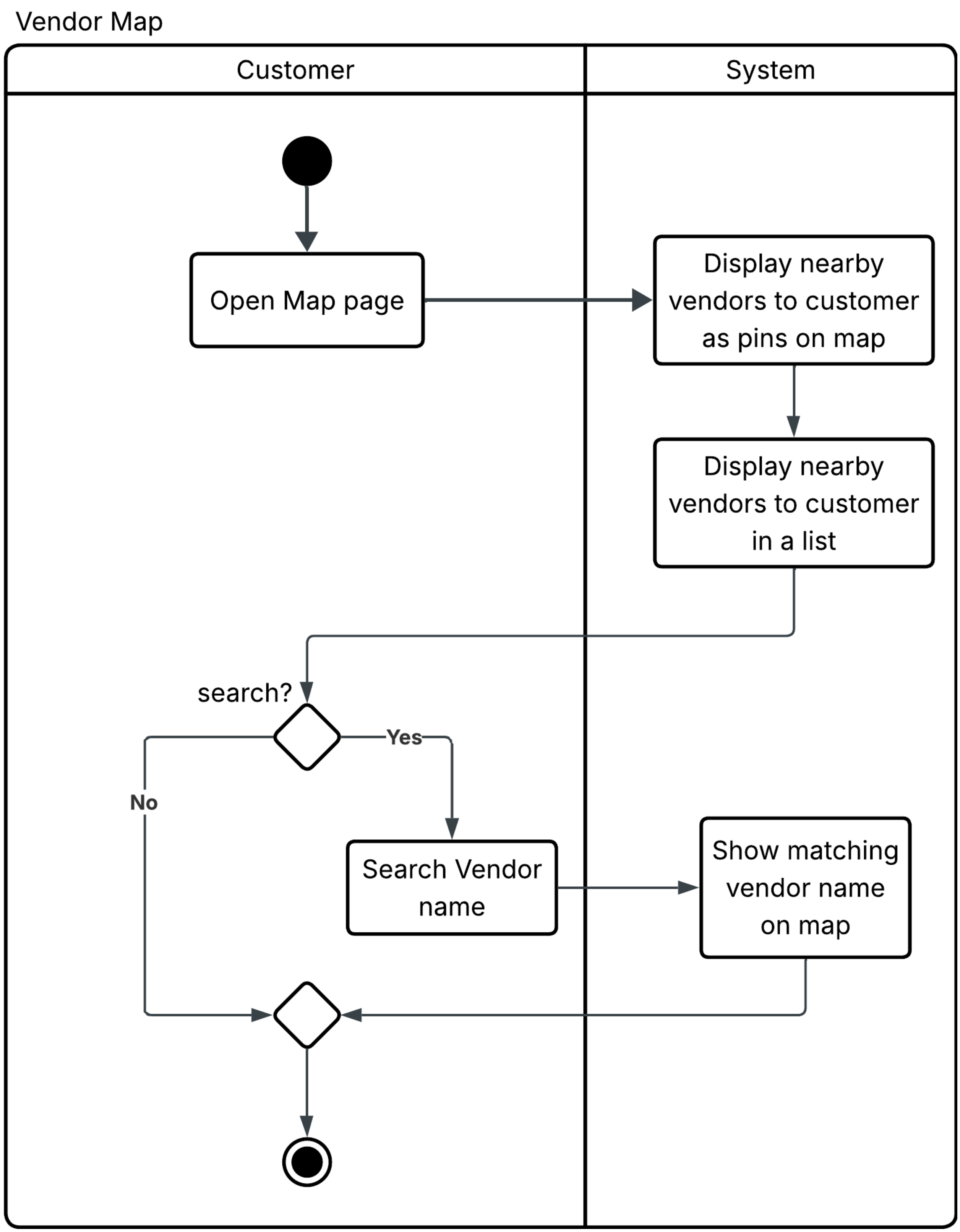
5. Process View

- Customer

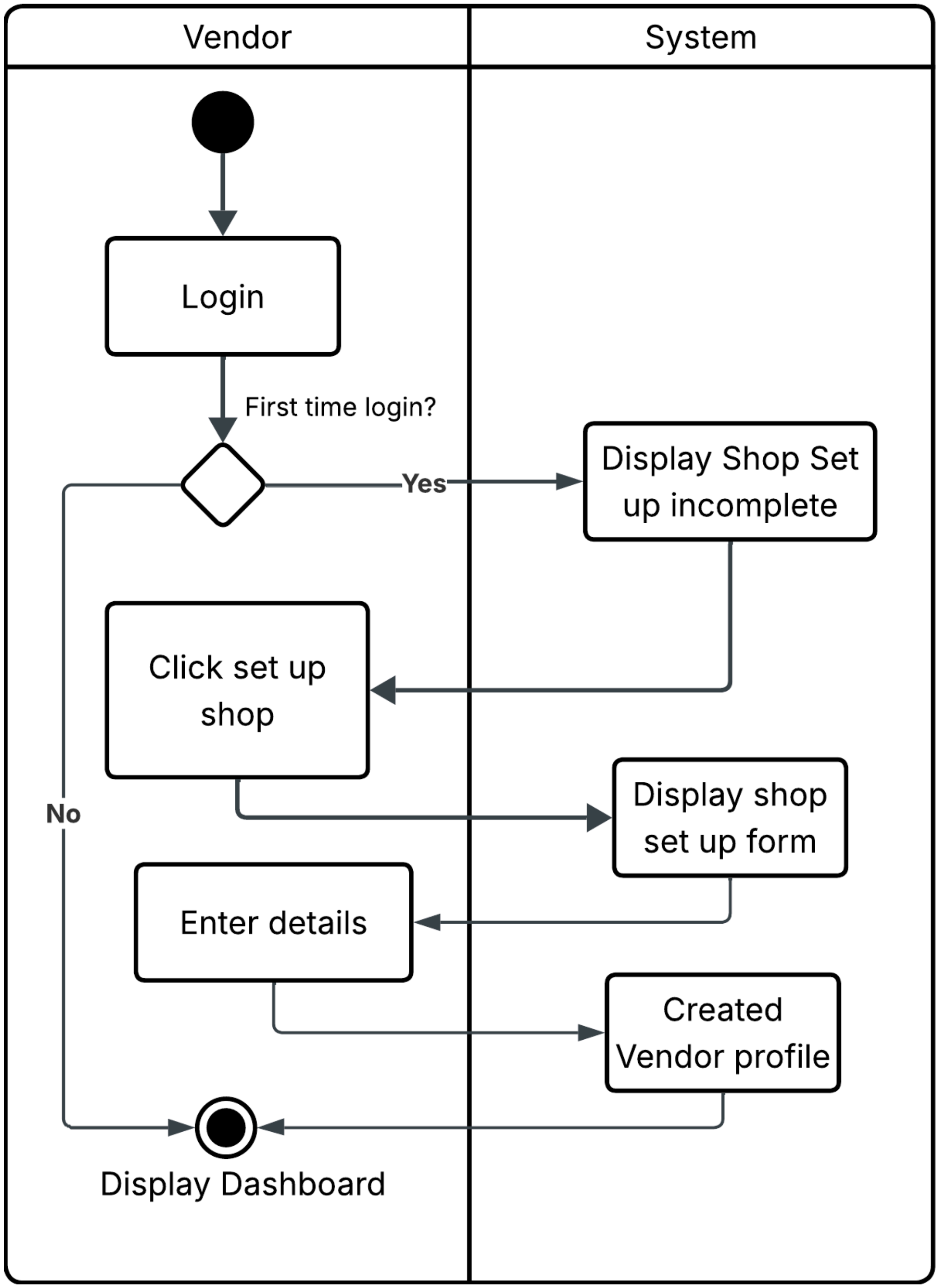

User Login
Customer
System
Google
Open system
Display Login with Google
Click login with google
Show Google Accounts
Choose google account
Authenticate
if success
First time login?
No
Yes
Create Account
Logged in

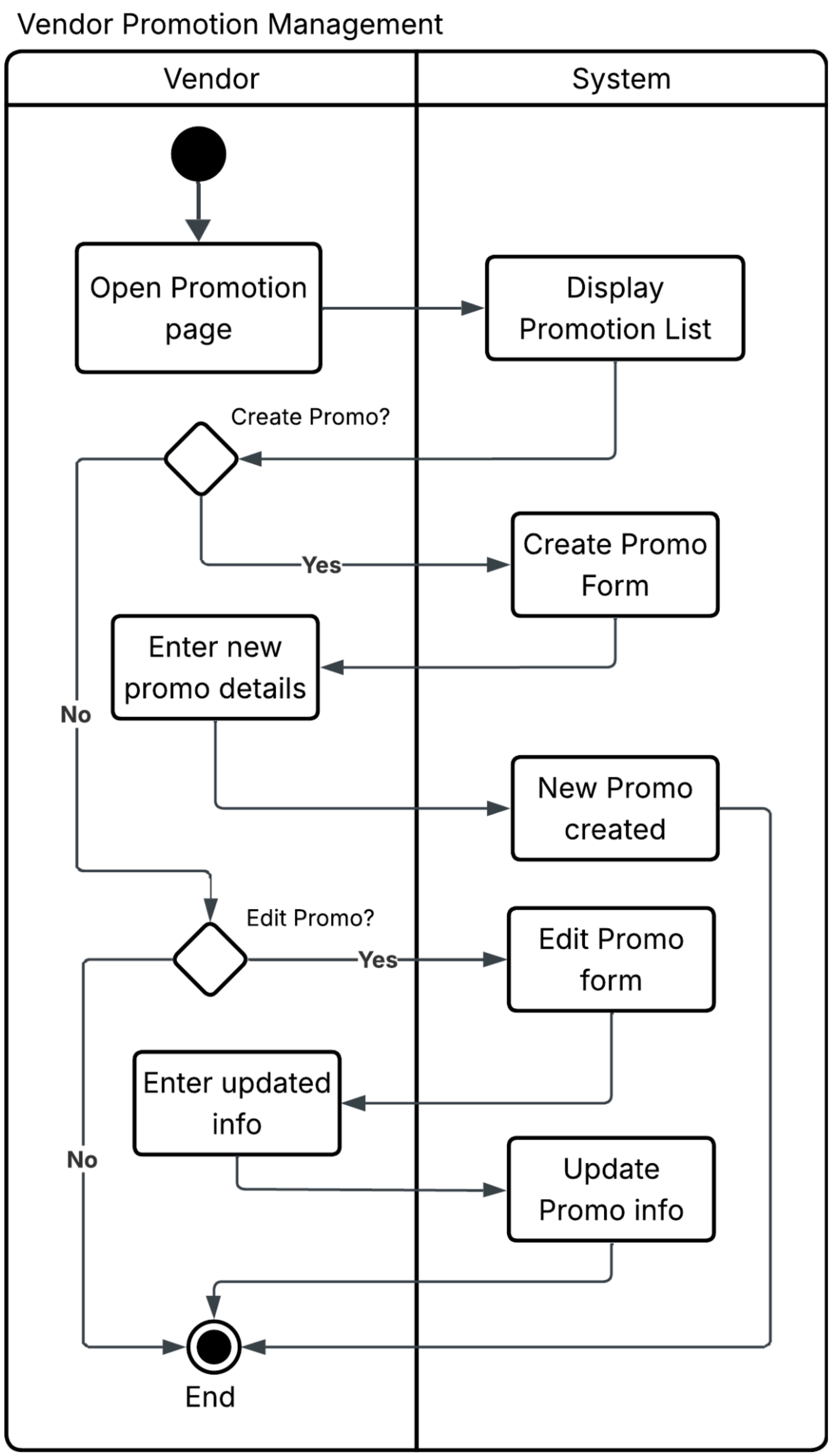
Profile

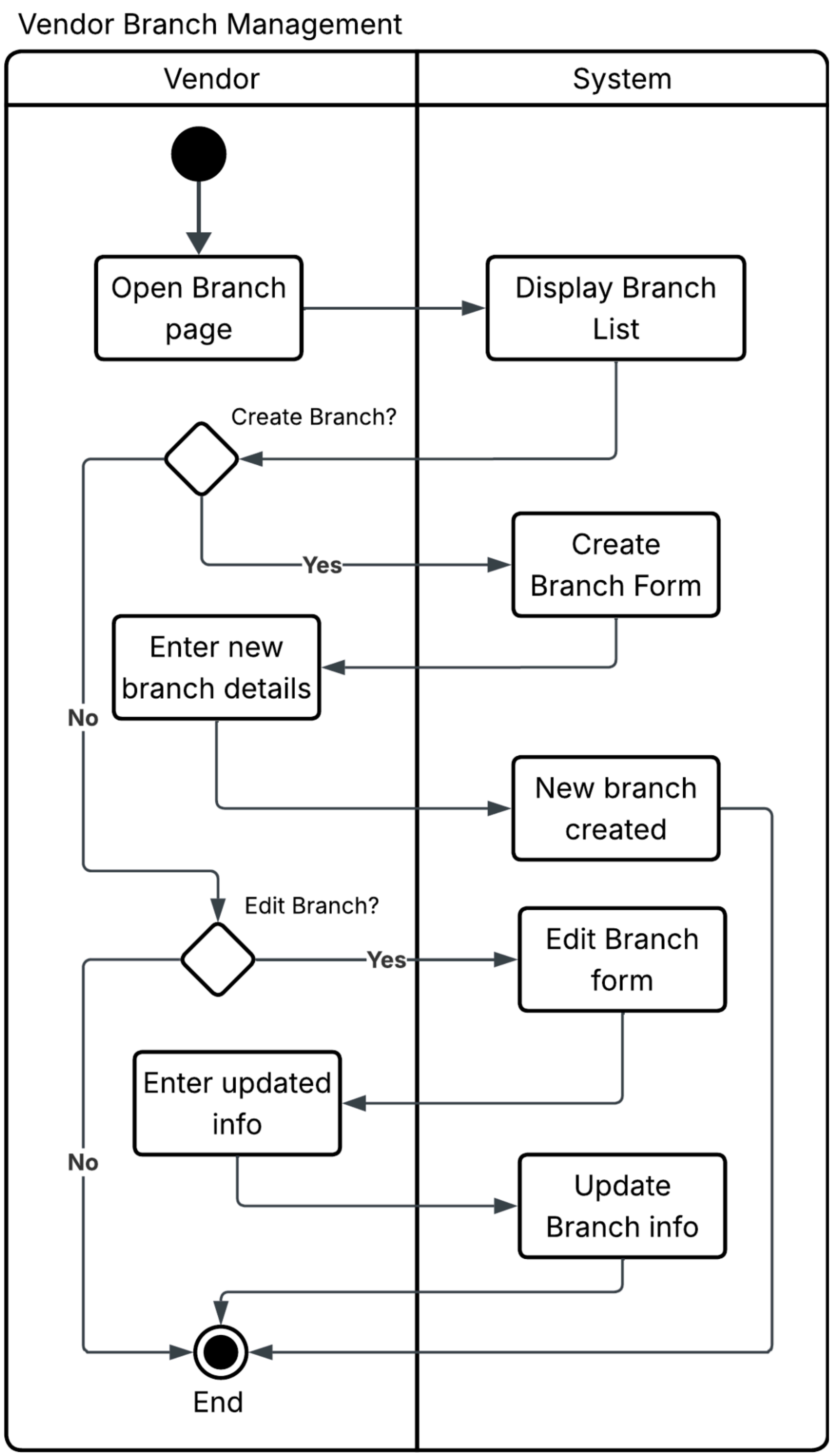

Customer
System
Open Profile page
Display customer information
edit?
Yes
No
Click edit info
Show edit info form


Vendor Map
Customer
System
Open Map page
Display nearby vendors to customer as pins on map
Display nearby vendors to customer in a list
search?
Yes
No
Search Vendor name
Show matching vendor name on map

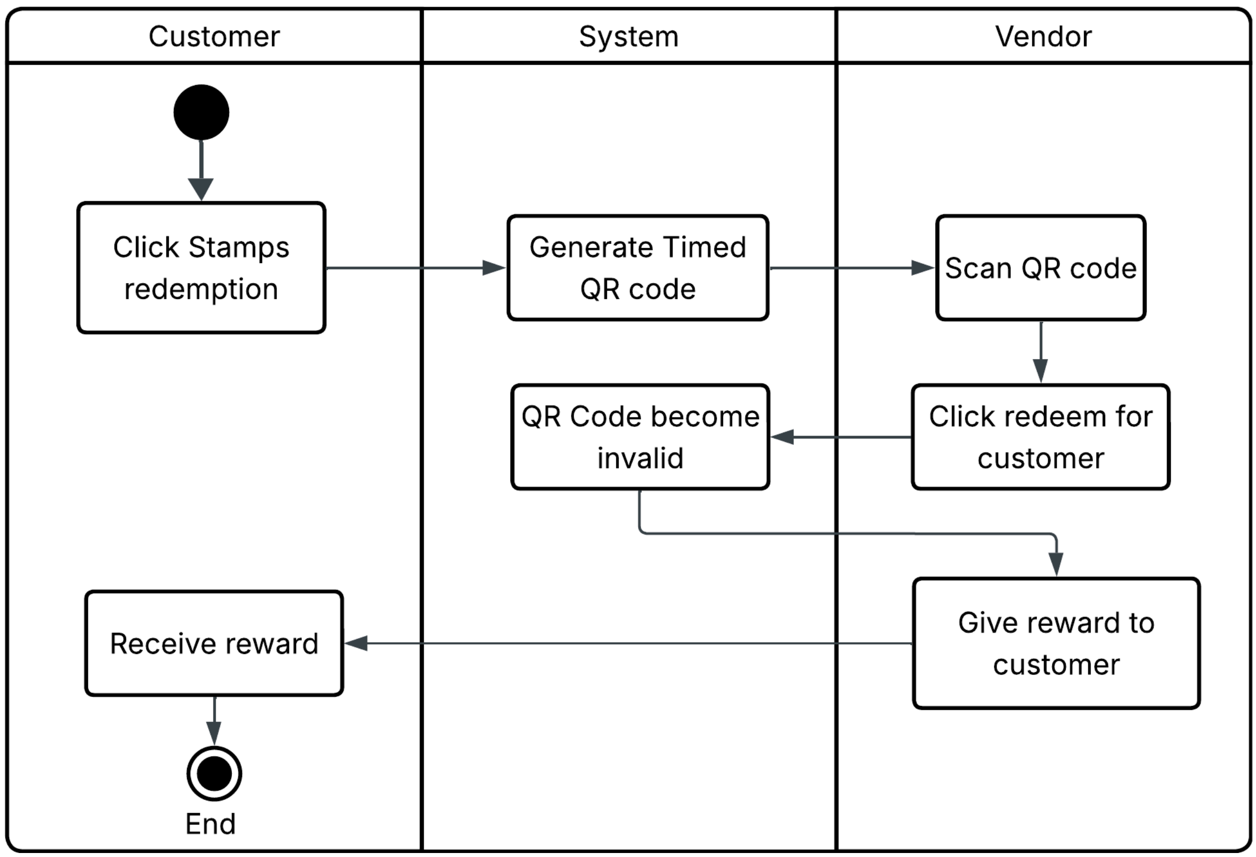
Vendor Set up



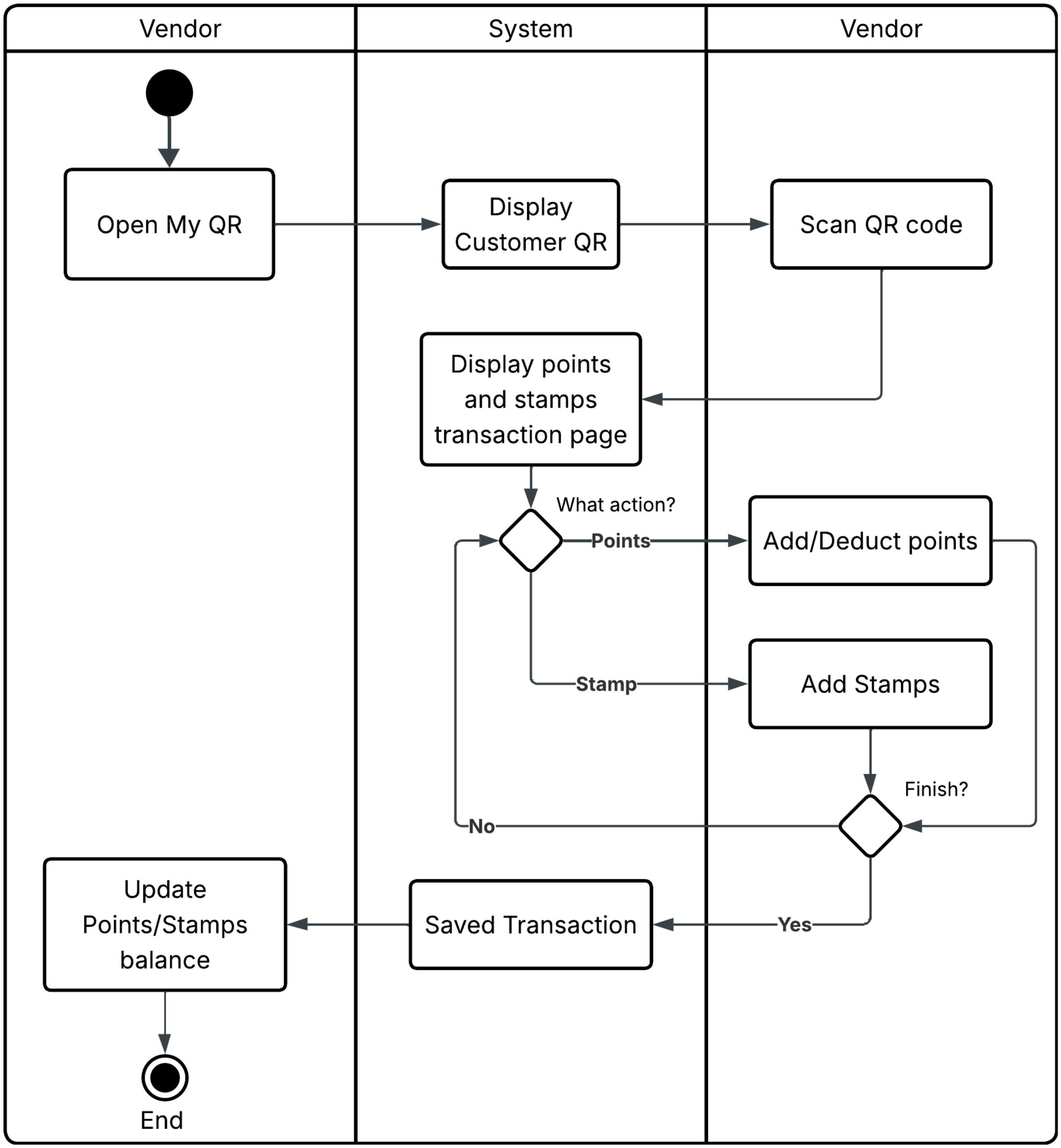

Vendor Promotion Management
Vendor
System
Open Promotion page
Display Promotion List
Create Promo?
Yes
No
Create Promo Form
Enter new promo details
New Promo created
Edit Promo?
Yes
No
Edit Promo form
Enter updated info
Update Promo info
End


Vendor Branch Management
Vendor
System
Open Branch page
Display Branch List
Create Branch?
Yes
Create Branch Form
Enter new branch details
No
New branch created
Edit Branch?
Yes
Edit Branch form
Enter updated info
No
Update Branch info
End

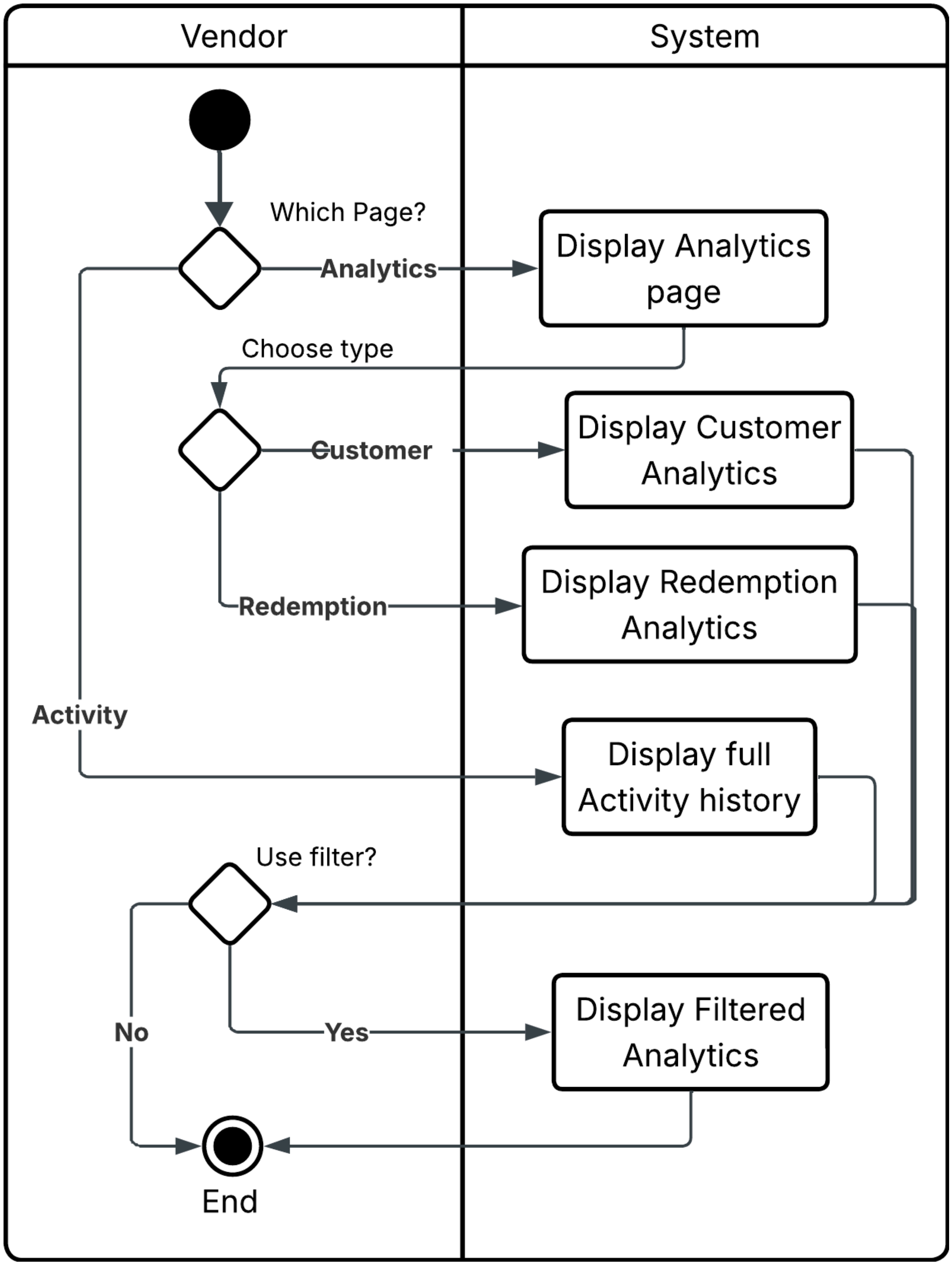
Customer Stamp Redemption

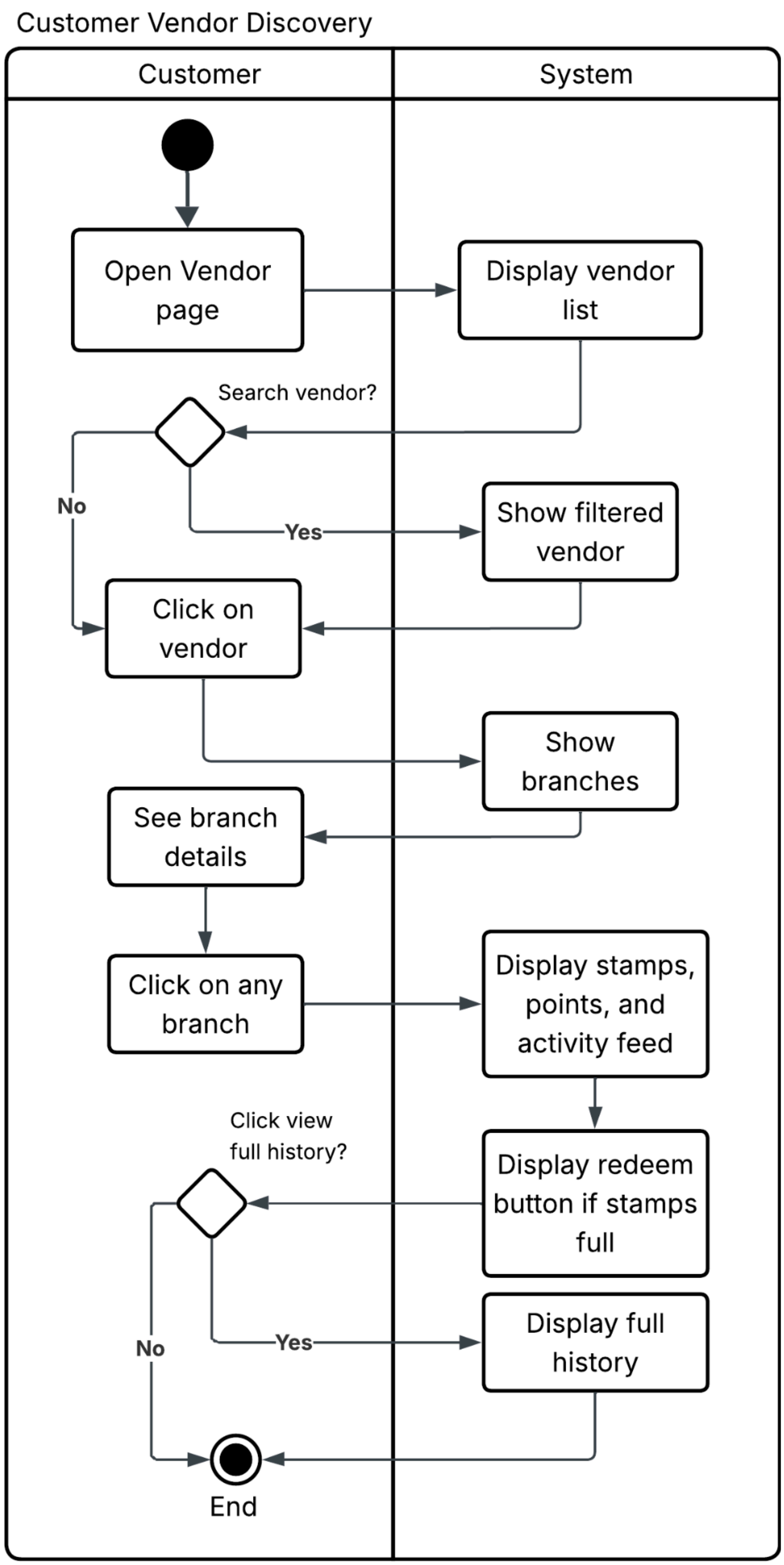


Customer Points Transaction



Vendor Analytics & Activity

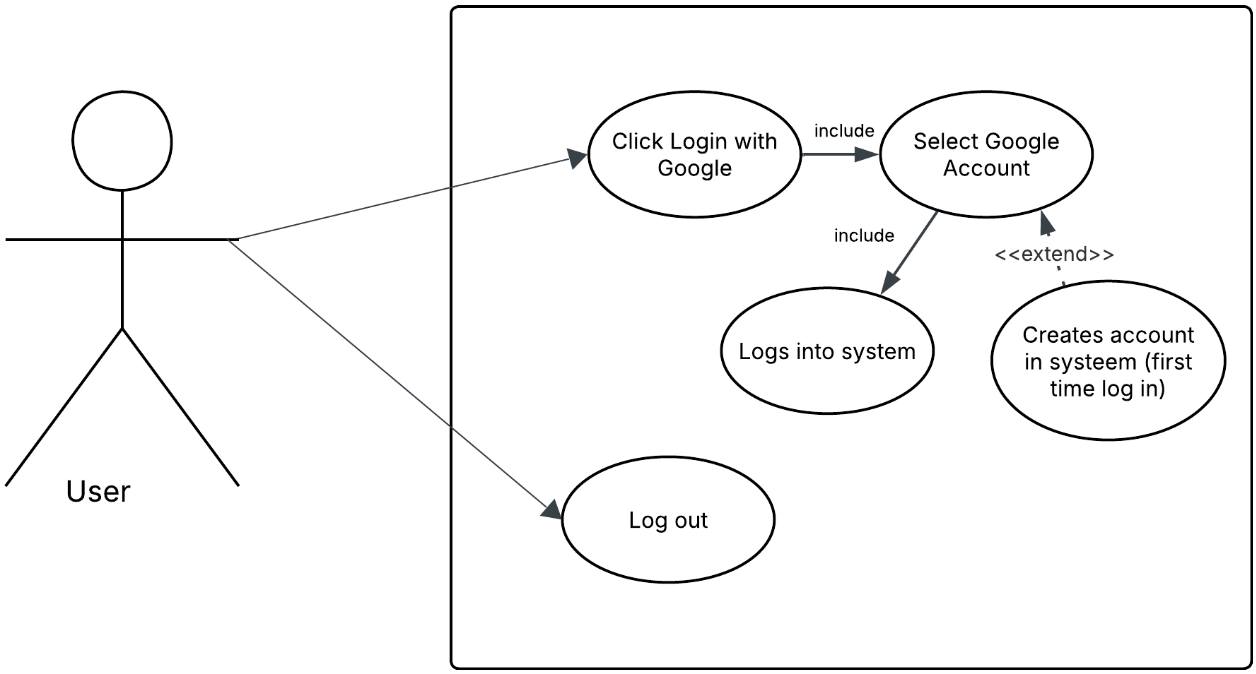


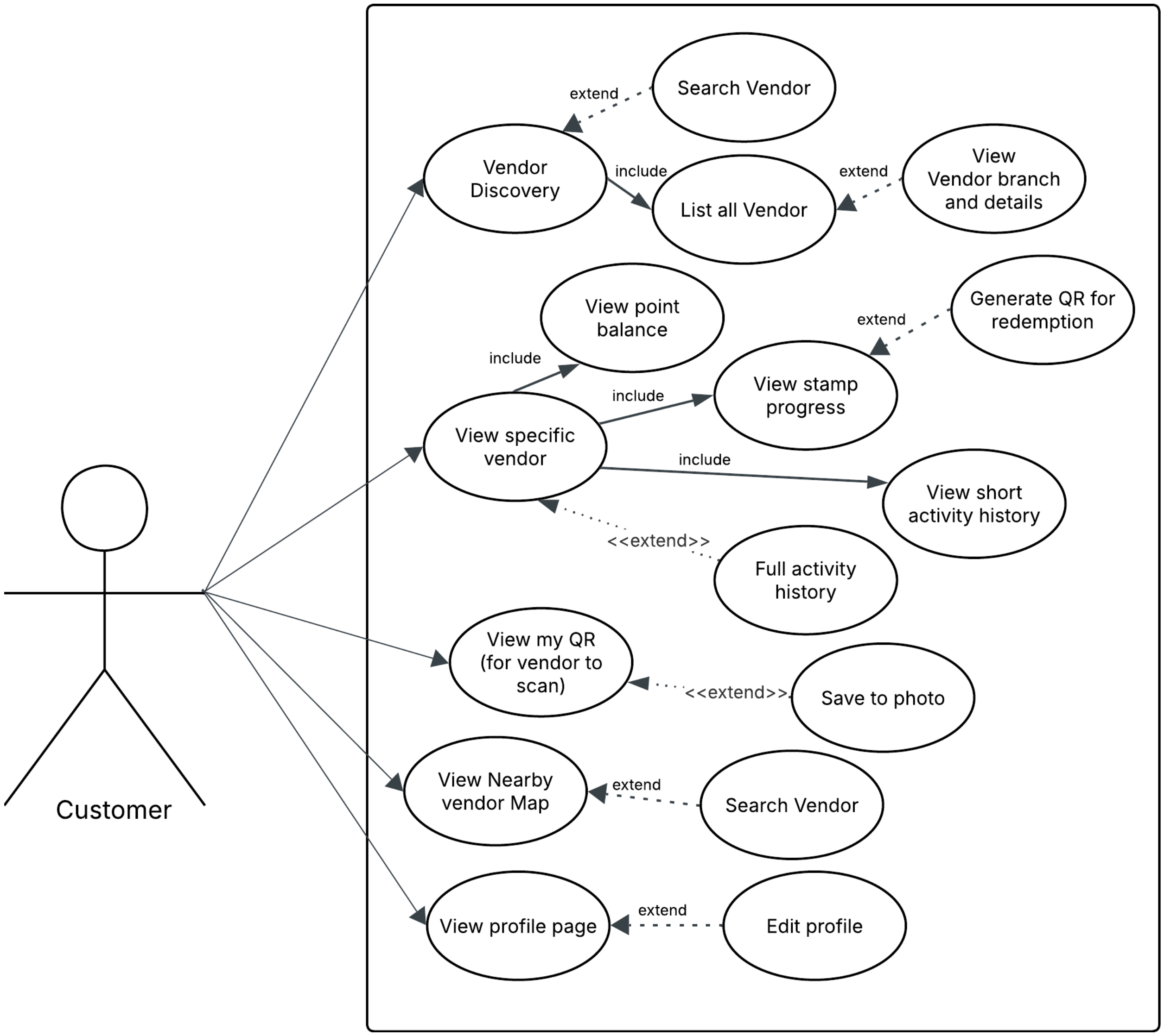

Customer Vendor Discovery
Customer
System
Open Vendor page
Display vendor list
Search vendor?
No
Yes
Show filtered vendor
Click on vendor
Show branches
See branch details
Click on any branch
Display stamps, points, and activity feed
Display redeem button if stamps full
Click view full history?
Yes
No
Display full history
End

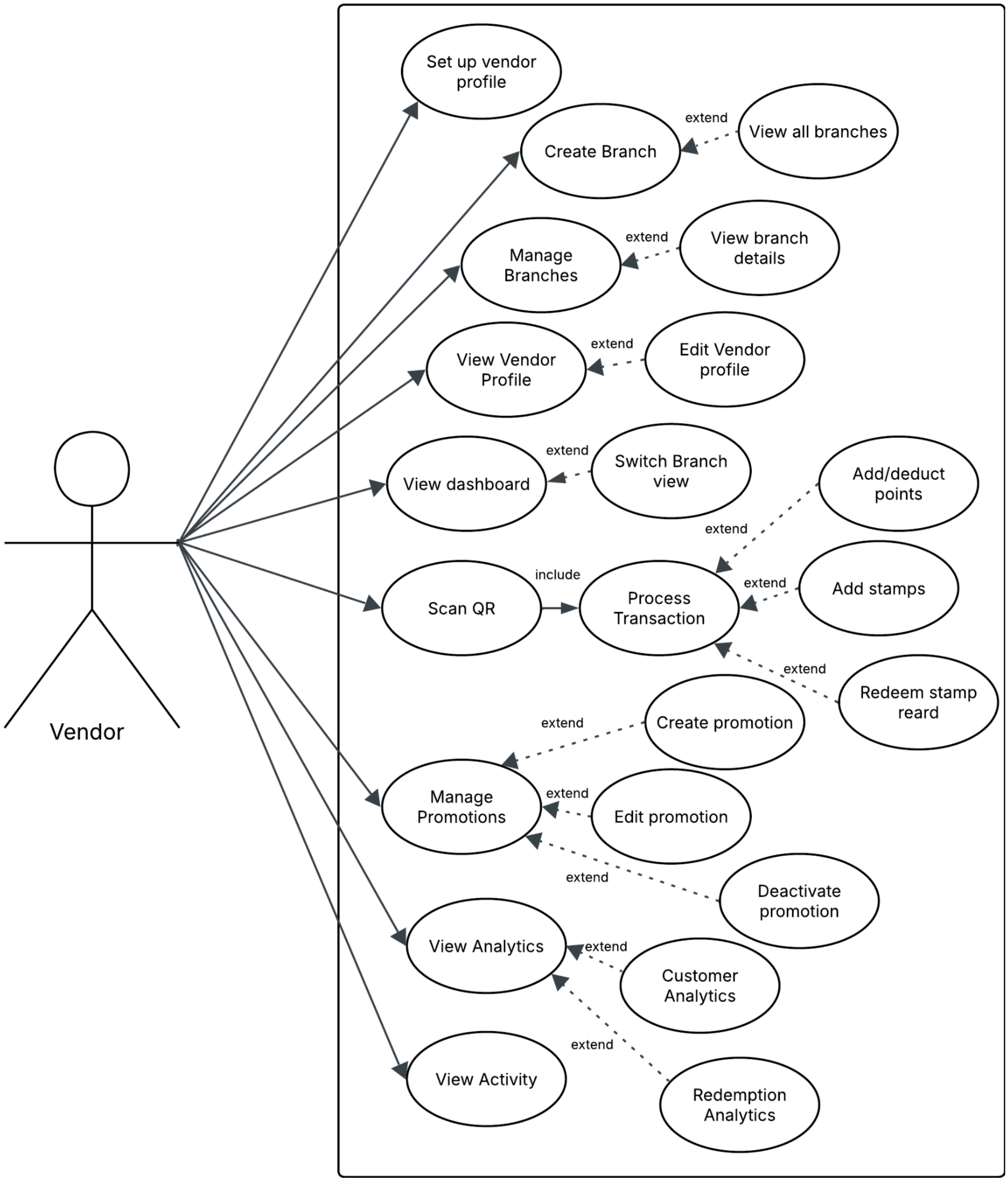
The Process View illustrates the interactions between users and system components during key business processes, demonstrating how requests are processed to complete platform operations.

6. Use case View

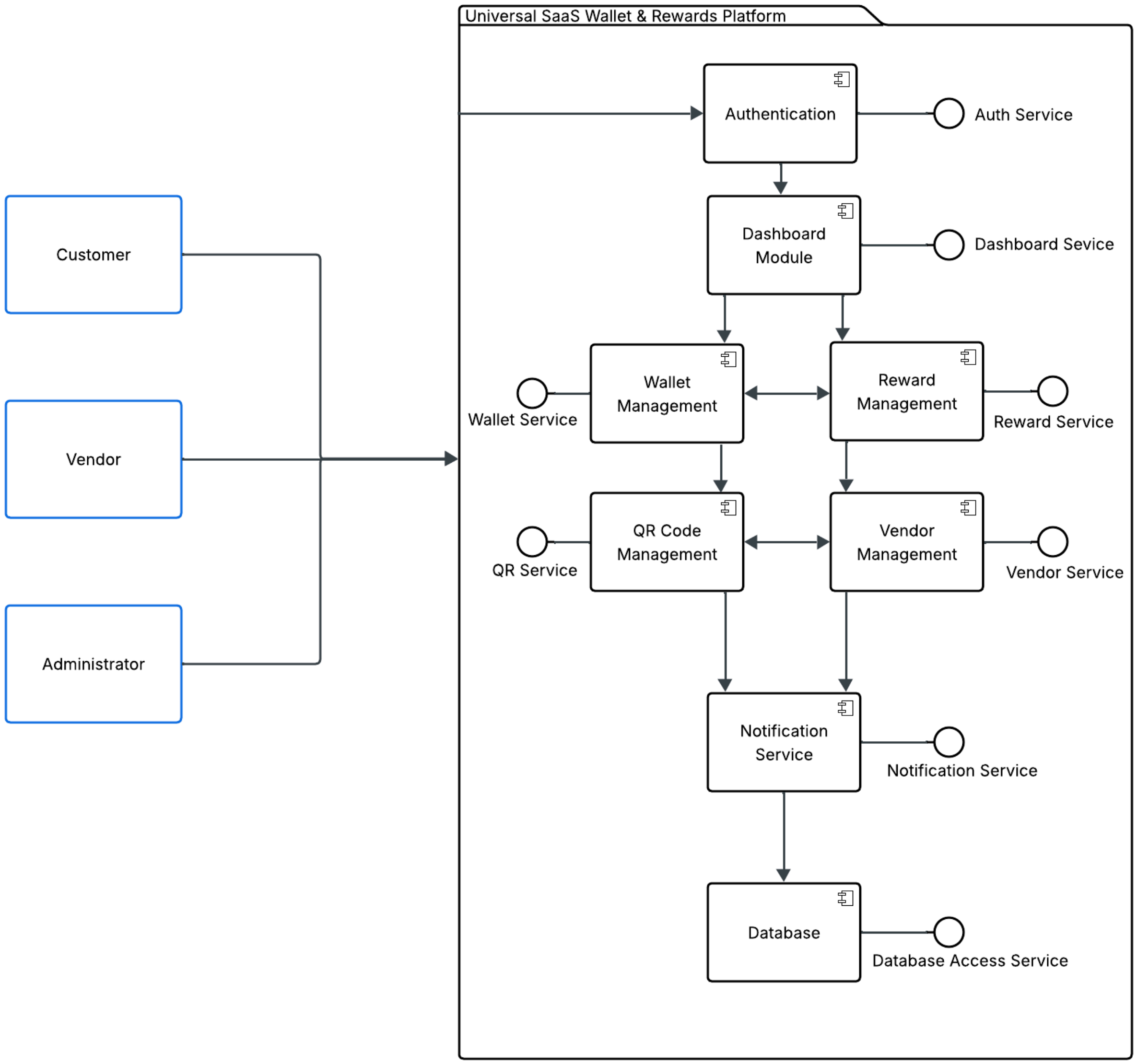
The Use Case View identifies the primary actors and their interactions with the Universal SaaS Wallet & Rewards Platform. It provides an overview of the system's major functionalities from the perspective of Customers and Vendors.




Customer
Search Vendor
extend
Vendor Discovery
include
List all Vendor
extend
View Vendor branch and details
View point balance
Generate QR for redemption
extend
include
include
View stamp progress
View specific vendor
include
View short activity history
<<extend>>
Full activity history
View my QR (for vendor to scan)
<<extend>>
Save to photo
View Nearby vendor Map
extend
Search Vendor
View profile page
extend
Edit profile

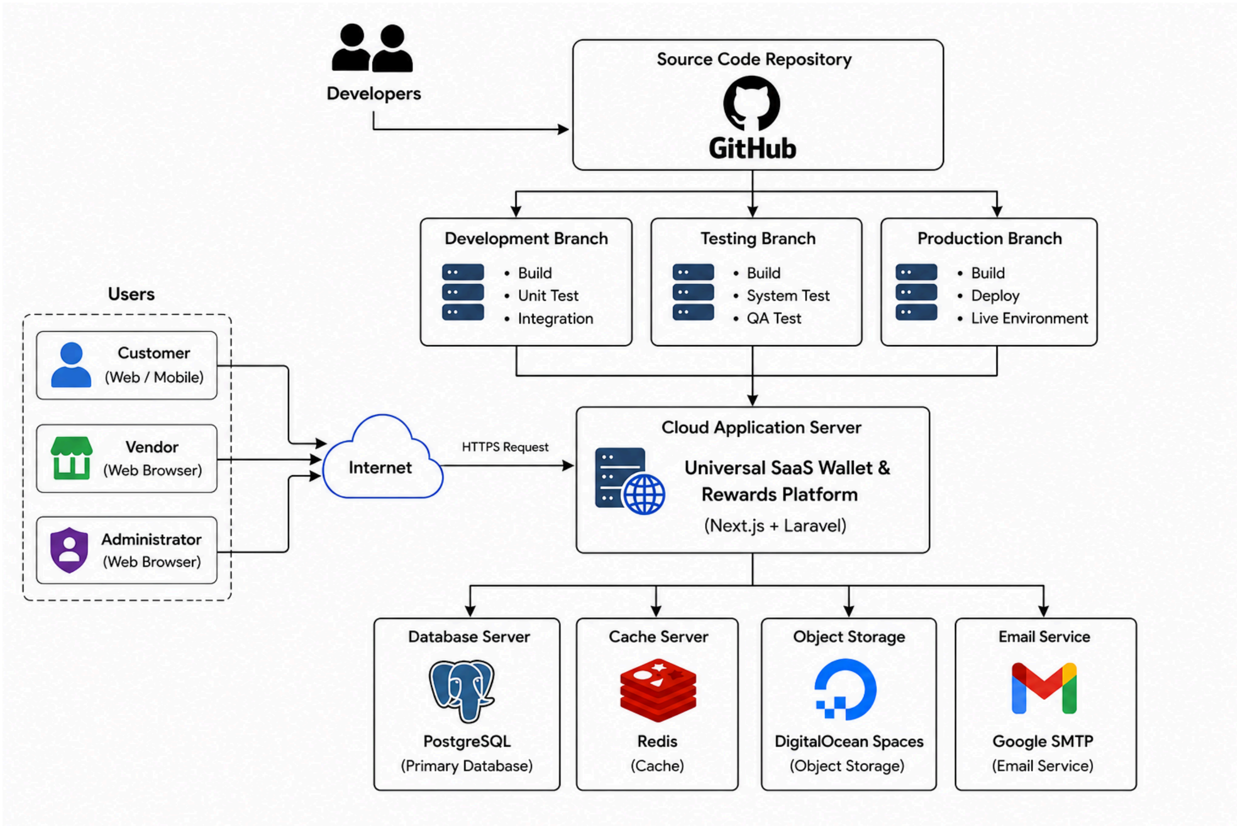

Vendor
Set up vendor profile
Create Branch
extend
View all branches
Manage Branches
extend
View branch details
View Vendor Profile
extend
Edit Vendor profile
View dashboard
extend
Switch Branch view
Add/deduct points
extend
Scan QR
include
Process Transaction
extend
Add stamps
extend
Redeem stamp reard
extend
Create promotion
Manage Promotions
extend
Edit promotion
extend
Deactivate promotion
View Analytics
extend
Customer Analytics
extend
View Activity
Redemption Analytics

7. Development View



The Development View illustrates the modular structure of the Universal SaaS Wallet & Rewards Platform. The system is organized into independent modules, including Authentication, Dashboard, Wallet Management, Reward Management, QR Code Management, Vendor Management, Notification Service, and the PostgreSQL Database. Each module is responsible for a specific business function while working together to support the platform's operations. This modular architecture improves maintainability, scalability, and simplifies future development.

8. Physical View



The Physical View illustrates how the Universal SaaS Wallet & Rewards Platform is deployed across the cloud infrastructure. Customers, Vendors, and Administrators access the platform through the Internet, where requests are processed by the cloud application server. The system communicates with PostgreSQL for persistent data storage, Redis for caching, DigitalOcean Spaces for object storage, and Google SMTP for email services. This deployment architecture ensures secure, scalable, and reliable platform operation.

9. User Stories

User Types	Epic	User Story	Acceptance Criteria
Customer	Authentication & Onboarding	**As a customer,** I want to log in exclusively using my Google account so that I can quickly and securely access my wallet without managing a new password.	- The system features a prominent "Continue with Google" onboarding action. - The system utilizes Google Sign-In OAuth to authenticate users. - First-time users are cleanly provisioned an account and a new universal wallet automatically upon selecting their Google profile. - Returning users are routed immediately to the main vendor listings interface.
	Vendor Directory (Coupons Home)	**As a customer,** I want to view a centralized listing of all participating vendors so that I can easily discover available loyalty programs.	- The system sets the initial landings screen ("Coupons") to list all vendors participating in the platform ecosystem. - Each listing clearly displays the brand identity (e.g., logo, business name, and quick program summaries). - A persistent global search field allows customers find their favorite stores. - Tapping an individual vendor tile opens the list of available branches for that selected vendor if they have multiple branches.

	Branch Selection & Loyalty Overview	As a customer, I want to select a branch of a vendor and view my loyalty points, stamp card progress, and recent activities so that I can continue collecting rewards across all branches of the same vendor.	- After selecting a vendor, the system displays all available branches of that vendor. - Each branch displays its branch name, address, operating hours, and contact information. - Customers can select any branch to continue. - All branches under the same vendor share the same loyalty points, stamp card progress, and reward redemption. - The loyalty page displays the customer's current point balance and stamp card progress. - A shortened activity feed highlights recent loyalty transactions. - A **View All History** action opens the complete transaction history.
	Reward Claiming & Verification	**As a customer,** I want to generate a timed redemption QR code once my stamp card is full so that I can securely claim my reward at the store.	- A "Claim Reward" actionable trigger becomes active only when a stamp card criteria is met (e.g., 10 / 10 stamps). - Initiating a claim creates a distinct, scannable transactional QR code containing specific reward metadata. - The redemption interface features an active visual countdown timer (e.g., "Code expires in 4:59") to secure the transaction. - The system invalidates the QR code automatically once used or when the timer hits

			zero to prevent fraud.
	Comprehensive History Ledger	**As a customer,** I want to view a complete historical timeline of my points and stamps so that I can audit past earned or spent actions.	- The interface displays a continuous, read-only chronological list of all account changes for the selected vendor's shared loyalty program. - The ledger clearly identifies the entry type (e.g., "Points Earned", "Stamp Earned", or "Reward Redeemed"). - Each entry documents the exact value shift, the specific calendar date, and the operational timestamp.
	Personal Identity QR Code	**As a customer,** I want to display my permanent personal account QR code so that merchant staff can easily scan it to issue my points or stamps.	- A dedicated "My QR Code" view is accessible via a fixed navigation tab item. - The view displays a permanent account QR code along with a clearly visible numeric Customer ID (e.g., CUS-000125). - A "Save to Photos" function allows users to save a local copy of their identification card to their device storage.

	Geolocation Discovery	**As a customer,** I want to view a map of nearby stores so that I can find participating vendor branches in my current area.	- The platform embeds an active map system displaying physical points-of-interest for registered vendor branches. - Search filters allow customers to screen locations by industry niches (e.g., Coffee, Dining, Retail). - A "Nearby Stores" card stack displays localized branches, complete with approximate distance calculations (e.g., 0.3 km away).
	Profile View	**As a customer,** I want to view my profile details and manage my session so that I can keep track of my member lifecycle.	- The profile panel surfaces basic Google-linked data including profile image, full display name, and email address. - The UI clearly showcases a milestone indicator stating when the customer joined the platform (e.g., "Member Since June 23, 2026"). - Features explicit actions for editing details or choosing to completely "Log Out" of the universal wallet app.

Vendor	Authentication & Shop Setup	As a vendor, I want to sign in using my Google account and manage my business and all of its branches from a single account so that I can easily operate my loyalty program.	- The system provides a **Continue with Google** login option. - First-time vendors are redirected to the **Shop Setup** page after signing in. - Vendors can upload a

			business logo and enter their business information. - After saving the profile, the vendor is redirected to the dashboard. - Returning vendors are taken directly to the dashboard. - Vendors can create multiple business branches using the same Google account. - Newly created branches are automatically linked to the same business account.
	Business Profile Management	As a vendor, I want to manage my business profile so that customers can view accurate and up-to-date business information.	- Vendors can view their business profile. - Vendors can update the business logo, business name, category, phone number, address, and website. - Changes are saved and reflected throughout the system. - Vendors can log out from the profile page.
	Dashboard Overview	As a vendor, I want to view my loyalty program summary and switch between branches so that I can monitor individual branch performance or my overall business performance.	- The dashboard displays today's overview, including Stamps Added, Points Added, Stamps Redeemed, and Points Deducted. - The dashboard provides quick access to Scan QR, Promotion, Branches, Analytics, and Activity. - If the shop profile has not been completed, the dashboard prompts the vendor to complete the

			setup before using loyalty features. - The dashboard provides a branch selector for switching between **All Branches** and individual branches. - Selecting an individual branch displays only that branch's statistics.
	Promotion Management	As a vendor, I want to create and manage loyalty promotions so that customers can earn points and collect stamps from my business.	- Vendors can create a new promotion. - Vendors can define the promotion name, reward, required stamps, points awarded, and promotion duration. - Vendors can edit or deactivate existing promotions. - Active promotions are immediately available to customers. - Customers can view active stamp card promotions and their current point balance. - Point redemption options are communicated by the vendor during in-store transactions.
	Branch Management	As a vendor, I want to create and manage multiple branches under one business account so that all branches share the same loyalty program while maintaining separate branch information.	- Vendors can view all registered branches. - Vendors can view detailed information for each branch. - If only one branch exists, the system displays a single-location message. - All branches share the same loyalty program

			and customer rewards. - The current main branch is clearly identified. - Vendors can create new branches under the same business account. - New branches are automatically connected to the existing business account. - All connected branches share customer points, stamp cards, promotions, and reward redemption. - Each branch maintains its own address, phone number, and operating hours. - Vendors can switch between **All Branches** and individual branches from the dashboard.
	Customer QR Transactions	As a vendor, I want to scan a customer's QR code so that I can add points, add stamps, or deduct points during a transaction.	- Vendors can scan a customer's QR code using the device camera. - Customer information is displayed after a successful scan. - Vendors can switch between Stamp and Points transactions. - Vendors can add stamps, add points, or deduct points. - Every transaction is automatically recorded in the activity history.
	Analytics & Reporting	As a vendor, I want to view business analytics so that I can evaluate customer engagement and reward performance.	- Vendors can view total customers and redemption statistics. - The analytics page

			displays customer growth charts. - The system shows points and stamps transaction summaries. - Vendors can view top-performing promotions. - Analytics can be filtered by a selected time period.
	Activity History	As a vendor, I want to view recent customer transactions so that I can monitor loyalty activities performed at my business.	- Vendors can view a chronological list of customer transactions. - The activity log includes points added, points deducted, stamps added, and reward redemptions. - Each activity displays the customer name, transaction type, value, date, and time. - Vendors can filter activities by transaction type and date.

Appendix C - AI Use Declaration

1. Vitou Raksmey

Agent(s) used:ChatGPT (OpenAI)
Prompt(s) used:The following prompts were used during the development of this project documentation and diagrams:
"Describe how a stock management system work for a pharmacy."
"List the workflow of stock management."
"Explain the 4+1 view for stock management ."
"Suggest improvements to my system design."
How the AI generated content was used:
The AI tool was used to assist with the system workflow, to provide hints in designing and formatting to make our design faster and more logical.

2. Tineo Lim

Agent(s) used: ChatGPT (OpenAI)
Prompt(s) used: "Based on the notes of the interview, could you help rewrite the whole process of the pharmacy in an easy read manner, and could you help draft the flow of the bpmn pools and swimlanes."
"Could you help explain to me how a digitalized inventory management system is supposed to look like?"
"Could you help explain the difference between an inventory management system and a stock management system?"
How the AI generated content was used:
So I just asked AI to help refine my messy notes I wrote from the interview, and I wanted to see its idea of a bpmn of the business is acceptable and I can adjust it based on my understanding. I also asked it about inventory management and stock management to see which one we would choose which at first we chose the inventory but later on we just switch to stock for a smaller scope.

3. Sophana Phat

Agent(s) used: ChatGPT (OpenAI), Brave AI (Brave Software)
Prompt(s) used:
prompt 1:
In an SDA document, can the process view and the development view be worked on concurrently? If not, which should be worked on first?

prompt 2:
Can you help me understand what I need to do before working on the development view? Ask me any clarifying questions necessary about my project, we've completed the project charter along with BPMN diagrams analyzing their business process.

prompt 3:
1. The system is a stock management system for a low scale pharmacy business.
- The project name is smart pharmacy stock management system

- Problems:
  i. Inefficient Stock Tracking
    a. Medicine stock is checked manually through visual inspection.
    b. Expired medicines may go unnoticed.
    c. Stock-in and stock-out are not recorded consistently.
    d. Medicine movement is difficult to trace.
  ii. Lack of Centralized Stock Information
    a. No clear overview of available stock.
    b. Difficult to monitor stock movement.
    c. Medicine quantities are manually counted and recorded.
    d. Possibility of incorrect stock deduction.

Goals and Objectives:
The overall goal is to provide pharmacies with a digital system that improves stock accuracy and operational workflow by cutting down the number of operational processes in stock management. The system aims to implement:

2. The pharmacist (staff) is mainly the only person interacting with the system
3. Medicine Stock Management System:
   - Real-time stock tracking using barcode scanning for both stocking and selling processes.
   - Allow manual quantity input through the system when selling partial quantities (e.g., not a full box).
   - Provide a low-stock and out-of-stock monitoring dashboard.
   - Monitor expiration dates using stored medicine information.
   - Update stock after medicine is dispensed through the system.
4. The system platform will be a web application. This project is only for design, we're not expected to build the system.
5. The system is closer to a single application.
6. So far we have login/logout, medicine catalog, and alerts. In the BPMN, we have diagrams for check stock, stock in, checking quality after items are delivered, organizing medicine, handling medicine expiration. Do note that these BPMN were meant to model their current business operations (without any digital system solutions)

The course is called "Software Design" which I was told is sort of an extension/continuation of System Analysis. My lecturer requires UML diagrams specifically.

How the AI generated content was used:
AI was used in identifying potential key modules of the system and their dependencies, clarify UML notations conventions, and design comparisons of various other software architectures.

4. Tithnisa Chap

Agent(s) used: ChatGPT (OpenAI), Google Gemini.
Prompt(s) used:
- “How should the logical view be written for a Software Design and Architecture report?”
- “What components should be included in a high-level architecture diagram for a pharmacy inventory management system?”
- “How do I structure the physical view in the 4+1 architectural model?”

- “Can you help improve the explanation for the high-level system architecture section?”
- “What is the difference between logical view and physical view in software architecture?”
- “How can I organize the system components in a high-level component diagram?”

How the AI generated content was used:
The responses from the AI tools were used as references to help understand the concepts and improve the wording of some parts of the report. All content was reviewed and modified by the team to ensure it accurately represents our project and system design.

5. Phalborith Teang

Agent(s) used: Gemini (Google)
Prompt(s) used: “Instead of a parallel gateway, what other tools can I use to organize these sub-tasks better without having 4 lines come out of the (diamond) parallel gateway?”

How the AI generated content was used:
- AI helped translate the visual UI’s of the wireframe into standard organized BPMN 2.0. It helped me understand the wireframe better and keep it more organized.
- The BPMN was constructed manually with the help of AI to keep it accurate to the intended software goal.

6. Kibriyo Shomirzoeva

Agent(s) used:
ChatGPT (OpenAI)
Prompt(s) used:
- Help me identify the main functions of a smart pharmacy stock management system.
- Can you help me design and check my use case diagram for the smart pharmacy stock management system?
- Suggest improvements to my system design.

How the AI generated content was used:
AI was used to help identify the main system functions, review the use case diagram, and clarify the correct use of UML relationships such as include and extend. It was mainly used as a guide to improve the structure and logical flow of the design, while the final diagram and decisions were completed manually.